Is a Description Worth a Thousand Demonstrations? Measuring the Effective Sample Size of Task Descriptions in In-Context Learning

by

Trung Quan Nguyen

A thesis submitted in partial fulfillment of the requirements for the degree of

Master of Science

in

Statistics

Department of Mathematical and Statistical Sciences
University of Alberta

© Trung Quan Nguyen, 2026

Abstract

Prompts for in-context learning usually pair demonstrations with a task description, yet its theories mostly model the demonstrations alone. We ask how many demonstrations a description is worth: its effective sample size (ESS), the number of demonstrations that lower a learner's error as much as the description does. For the Bayes-optimal predictor in in-context linear regression, the ESS depends on the description's specificity (how much uncertainty about the task it removes) and its reliability (the probability that it is relevant). We prove that a reliable description's ESS grows linearly with the task dimension when the description is loose and with its precision when it is specific. A description of only part of the task is worth about the number of directions it covers, until demonstrations resolve the rest. Without demonstrations, a description that may be irrelevant is worth a bounded number of demonstrations however specific it is. With demonstrations, trusting it fully can be worse than ignoring it. Meta-trained Transformers come close to the Bayes-optimal ESS and learn how much to trust a description from training. Pretrained language models answering without reasoning respond to a description's stated specificity far less than the Bayes-optimal predictor does, even when nothing else changes, and they do not adjust to its stated reliability in proportion. A description worth 36 demonstrations to that predictor is worth 1 to 8 to these models. Models that reason first follow the stated specificity, and two come close to the Bayes-optimal ESS.

Preface

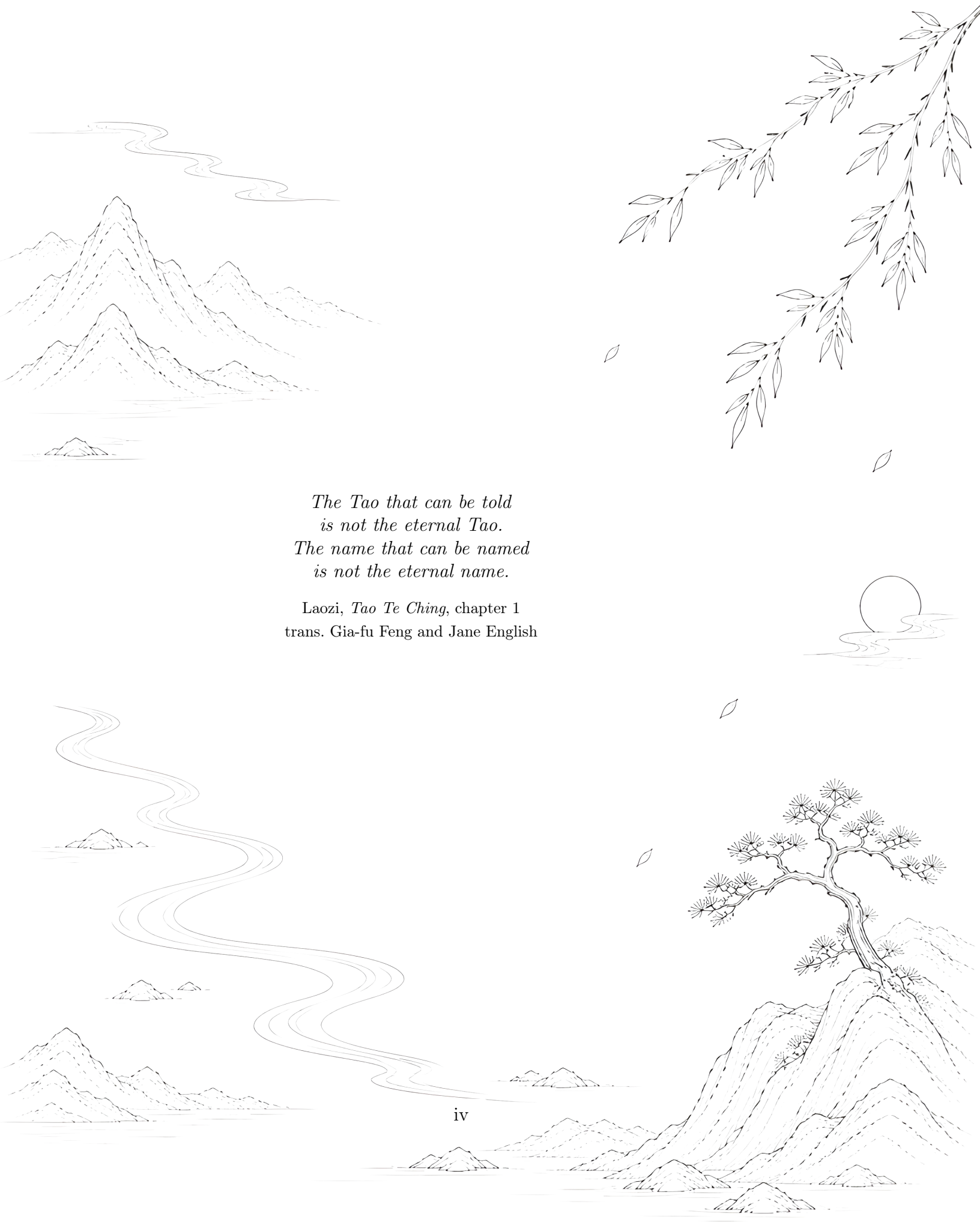
This thesis is an original work by Trung Quan Nguyen, carried out under the supervision of Professor Linglong Kong and Professor Bei Jiang. **[TODO: Describe any contribution of the supervisors or others to the research questions, design, analysis, or writing, or state that there was none.]**

The thesis is based on the manuscript "Is a Description Worth a Thousand Demonstrations? Measuring the Effective Sample Size of Task Descriptions in In-Context Learning" by Bruce Quan Nguyen, the name under which the author writes, prepared in the format of the International Conference on Machine Learning. Chapters 1 and 3 to 7, Chapter 8 except Section 8.2, and Appendix A present that manuscript's material, reorganized for a thesis. Chapter 2 is new, except that Sections 2.2, 2.5, 2.6, 2.7, and 2.8 draw on the manuscript's review of related work. Section 8.2 is also new. **[TODO: State whether the manuscript has been submitted or published; if so, give the full citation.]**

The research involved no human participants, animals, or personal data, so no research ethics approval was required.

Large language models are the object of study in Chapter 7. Separately, generative AI assistants were used in carrying out this research. **[TODO: Confirm and adjust this draft: Anthropic's Claude Code and OpenAI's Codex were used to write and run experiment code, to check derivations, and to draft and edit text; the author designed the study and reviewed, and takes responsibility for, all code, results, and text. Confirm that the program and supervisory committee permitted this use.]**

[TODO: Acknowledge third-party funding, such as a supervisor's grant, a scholarship, or Decaude Labs, and any paid compute; or state that the work was unfunded.]



The Tao that can be told
is not the eternal Tao.
The name that can be named
is not the eternal name.

Laozi, *Tao Te Ching*, chapter 1
trans. Gia-fu Feng and Jane English

Acknowledgements

I thank my supervisors, Professor Linglong Kong and Professor Bei Jiang, for their guidance and support throughout my program. **[TODO: Thank the examining committee, colleagues, funders, and family as you wish.]**

Table of Contents

Abstract ii

Preface iii

Quotation iv

Acknowledgements v

List of Tables ix

List of Figures x

1 Introduction 1

1.1 Contributions . . . 3

1.2 Outline . . . 4

2 Background 6

2.1 In-Context Learning . . . 6

2.2 In-Context Learning as Bayesian Inference . . . 7

2.3 In-Context Linear Regression . . . 8

2.4 Squared Error . . . 8

2.5 The Effective Sample Size of a Prior . . . 10

2.6 Robust Mixture Priors . . . 12

2.7 Descriptions in Theories of ICL . . . 12

2.8 Descriptions and Priors in Pretrained Language Models . . . 13

3 Problem Setup 14

3.1 Tasks and Demonstrations . . . 14

3.2 Descriptions . . . 14
3.3 Error and Effective Sample Size . . . 15

4 Specificity: Reliable Descriptions 17
4.1 The ESS of a Reliable Description . . . 17
4.2 Descriptions of Part of the Task . . . 18
4.3 Proof of Theorem 4.1 . . . 20
4.4 Extra Precision with Many Demonstrations . . . 22
4.5 Proof of Proposition 4.2 . . . 22

5 Reliability: Descriptions That May Be Irrelevant 24
5.1 The Price of Not Knowing Relevance . . . 24
5.2 The ESS of a Description That May Be Irrelevant . . . 25
5.3 Full Trust Can Be Worse Than No Description . . . 26
5.4 Proofs of Proposition 5.1 and Corollary 5.2 . . . 27
5.5 Derivation of Equation (5.3) . . . 28

6 Meta-Trained Transformers 30
6.1 Setup . . . 30
6.2 Results . . . 31
6.3 Training Details . . . 34

7 Pretrained Language Models 36
7.1 Setting . . . 36
7.2 The Bayes-Optimal Predictor . . . 37
7.3 Models . . . 38
7.4 Measurement . . . 39
7.5 Results . . . 40
7.6 Further Analyses . . . 45
7.6.1 Specificity Control . . . 45
7.6.2 Readings of the Error Curves . . . 46
7.6.3 Written Readout . . . 48
7.6.4 Actual Unreliability . . . 49
7.6.5 Wording of the Description . . . 50
7.6.6 Reasoning . . . 51

7.6.7 A Frontier Model Inside an Agent Harness 53

8 Conclusion **56**

8.1 Limitations 56

8.2 Future Work 57

Bibliography **59**

A Language-Model Results at Every Number of Demonstrations **63**

List of Tables

4.1 The ESS of a partial description 20

6.1 The meta-trained networks against the Bayes-optimal predictors . . . 35

7.1 The ESS and the weight on the stated value for each model 41
7.2 The specificity control 47
7.3 The headline ESS under four readings of the error curve 48
7.4 The ESS under actual unreliability 50
7.5 The weight on the stated value under four wordings 51
7.6 Models that reason before answering 53
7.7 GPT-5.6-Luna's weight on the stated value 55
7.8 GPT-5.6-Luna's ESS 55

A.1 The ESS at every number of demonstrations 64
A.2 The ESS at every number of demonstrations, written readout 66
A.3 The weight on the stated value at every number of demonstrations . . 67

List of Figures

1.1 A description is worth a number of demonstrations 5

4.1 The ESS of a description against its specificity 19

6.1 The ESS of a description to meta-trained networks 32
6.2 The networks' ESS as demonstrations accumulate 33

7.1 The error curves from which the language-model ESS is read 42
7.2 The ESS of a stated prior to pretrained language models 43

Chapter 1

Introduction

Large language models are now directed mostly through text written in natural language. A user writes a prompt that describes a task, a developer writes a system prompt that sets the model's behaviour, and a coding-agent user writes instruction files and skills that tell the agent how to work in a project (Chatlatanagulchai et al., 2025). Writing this text remains a craft rather than a science. Practitioners learn what works by trial and error, and the published evidence on how much such text helps is mixed. In prompt-based fine-tuning, a prompt can be worth hundreds of training examples (Le Scao & Rush, 2021), and a written summary of a task can stand in for many demonstrations of it (Honda et al., 2025). Yet models learn as fast from irrelevant or misleading instructions as from instructive ones (Webson & Pavlick, 2022), and they let the evidence in a prompt outweigh an explicit instruction (Gupta et al., 2025). Coding agents given repository instruction files also do not generally solve more tasks (Gloaguen et al., 2026).

To study this text scientifically, we need two things. The first is a unit in which to measure what a piece of text is worth to a model. The second is a small set of properties of the text that determine that worth, so that different texts can be compared by their properties rather than one prompt at a time. In this thesis, we take a first step in the simplest case: text that tells a learner which task it faces. Brown et al. (2020) distinguished the natural-language *task description* at the start of a prompt from the *demonstrations*, the solved examples, that follow it. Following them, we call everything in a prompt other than its demonstrations a *description*. A description may be written as a statement about the task or as a command. Either way, it gives the learner information about which task it faces.

Theories of in-context learning (ICL) offer little guidance here. Most of them model the prompt as a sequence of demonstrations alone (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). Those that add a description show that a learner can use it to predict better (Huang & Ge, 2025), but they do not say how much better, or what a description is worth when it may be wrong.

Bayesian statistics already has both a unit and a set of properties for a closely related object, the prior. We therefore take the Bayesian view of ICL (Xie et al., 2022). In this view, the prompt is evidence about a latent task, and each demonstration is one noisy observation of that task. A description is a statement about which task generated the demonstrations, so it sharpens the learner's prior. We set aside instruction following. Bayesian statisticians measure the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information (Morita et al., 2008). The literature on informative priors also characterizes a prior by two properties, which we adopt as two aspects of a description.

The first aspect is *specificity*, the precision of the prior that the description gives (Morita et al., 2008; Neuenschwander et al., 2020). "Write this report like an accountant" only loosely indicates the task, whereas a description that fixes the report's sections, figures, and accounting standard nearly states it. The second aspect is *reliability*, the probability that the description is relevant to the task at hand. In Bayesian analysis, this probability is the weight of an informative prior in a robust mixture, which protects the analysis when the prior conflicts with the data (Schmidli et al., 2014; Evans & Moshonov, 2006). Descriptions can be unreliable for ordinary reasons. A description written for one report may be pasted into the prompt for another, or it may simply be wrong. Descriptions have other aspects as well. A description can be specific about some aspects of a task and silent about others, and Section 4.2 treats such partial descriptions. Section 8.2 discusses another aspect: descriptions that state the structure of a task rather than where it lies.

We define the ESS of a description in the same way as the ESS of a prior, as the number of demonstrations that the description can replace (Figure 1.1). We first compute it in noisy in-context linear regression with a Gaussian task prior (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where Bayes-optimal prediction has a closed form. We then measure it for Transformers meta-trained on this setting and for pretrained language models. An example from Chapter 7 illustrates what the unit measures. A language model is shown numbers drawn from a normal distribution

with standard deviation 60 and is asked to predict the next one. The prompt also contains the description "Its mean was drawn from a normal distribution with mean 510 and standard deviation 10." To a predictor that is Bayes-optimal given what the prompt states, this sentence is worth 36 demonstrations, the ratio of the two variances, $60^2/10^2$. To gpt-oss-20b answering without reasoning, it is worth 2.8 demonstrations (the median over one, two, and four demonstrations shown; Table 7.1). A control in the same chapter keeps the stated value and the demonstrations fixed and changes only the stated standard deviation. When the sentence states a standard deviation of 60 instead of 10, the Bayes-optimal ESS falls from 36 to 2.5, while gpt-oss-20b's ESS barely moves, from 2.8 to 2.7.

1.1 Contributions

Our contributions are as follows.

- **A common unit.** We define the ESS of a description as the number of demonstrations that lower a learner's error as much as the description does (Definition 3.1). The definition follows the data-dependent ESS of a prior (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021), which depends on the observations already in hand. It applies to any learner. With squared error as the measure of error, it requires only the learner's answers.

- **Theory.** For the Bayes-optimal predictor, a reliable description that leaves a fraction r of the prior variance is worth about $(1 - r)\, d$ demonstrations when it is loose, where d is the task dimension. When it is specific, it is worth $c+d+1$ at a high signal-to-noise ratio, where c is the precision of its prior measured in demonstrations (Theorem 4.1). The term c is the ESS that Morita et al. (2008) give the same prior, which matches information. Our ESS matches errors instead, and this adds a dimension term, which fades as demonstrations accumulate. A description of only k of the d directions is worth about k demonstrations when none are in hand, whatever its precision, and $(1 - r)\, c\, k/d$ once many are (Proposition 4.2). A description that may be irrelevant behaves differently. With no demonstrations in hand, it is worth a bounded number of demonstrations however specific it is, at most 15.6 at $d = 16$ and $a_0 = 10$ when it is relevant with probability 0.9. Once demonstrations are in hand, trusting it fully can be worse than ignoring it (Proposition 5.1, Corollary 5.2,

and Section 5.3). The bound follows from the law of total variance.

- **Experiments.** Meta-trained Transformers come close to the Bayes-optimal ESS, and they learn how much to trust a description from their training data (Chapter 6). Pretrained language models answering without reasoning adjust to neither a description's stated specificity nor its stated reliability as the Bayes-optimal predictor does. They respond to the stated specificity far less than the Bayes-optimal predictor even when nothing else in the prompt changes. A description worth 36 demonstrations to that predictor is worth a median of 1 to 8 demonstrations to these models. Models that reason before answering follow the stated specificity, and two of them, Claude Haiku 5.5 and GPT-5.6-Luna, come close to the Bayes-optimal ESS (Chapter 7).

1.2 Outline

The results of Chapters 3 to 7 also appear in the manuscript described in the Preface. This thesis presents them in full, together with the background they rest on, complete proofs, and all experiments. The rest of the thesis is organized as follows. Chapter 2 reviews the background: in-context learning and its interpretation as Bayesian inference, in-context linear regression, squared error, the effective sample size of a prior, and robust mixture priors, followed by the work most closely related to ours. Chapter 3 defines the setting, the model of a description, and the ESS of a description. Chapter 4 derives the Bayes-optimal ESS of a reliable description, both of the whole task and of only part of it. Chapter 5 derives the Bayes-optimal ESS of a description that may be irrelevant. Both chapters include complete proofs. Chapter 6 measures the ESS for meta-trained Transformers, with three training seeds for the specific description, and follows it as demonstrations accumulate. Chapter 7 measures the ESS for pretrained language models given a stated prior. It includes a control that changes only the stated specificity, and runs with models that reason before answering. Chapter 8 summarizes the findings, discusses their limitations, and outlines future work. Appendix A reports the language-model results at every number of demonstrations.

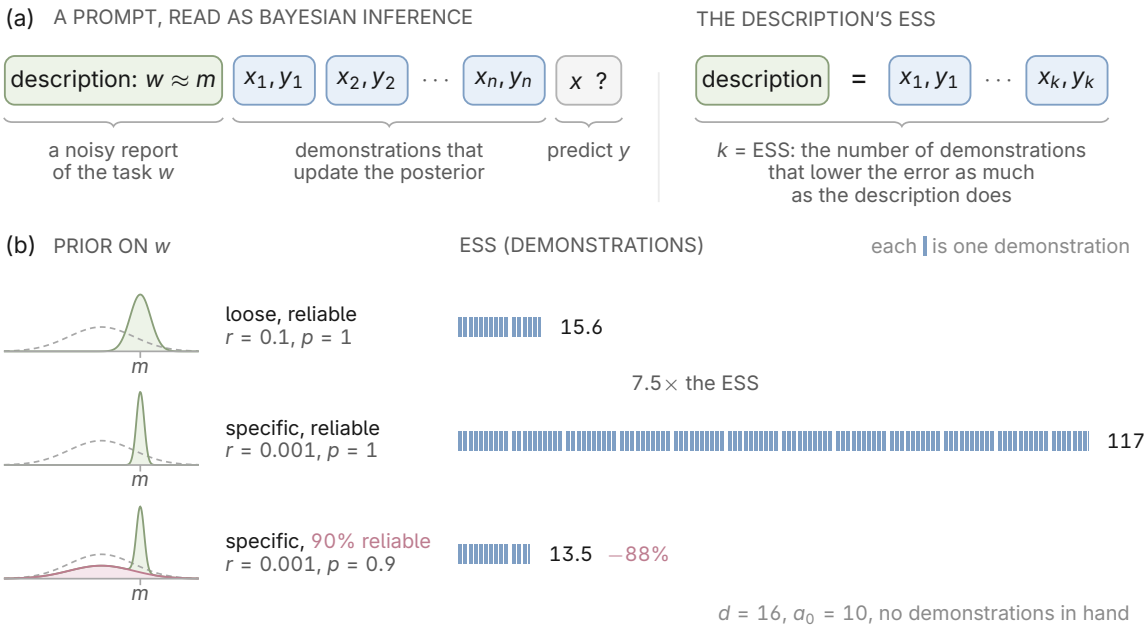


Figure 1.1: **A description is worth a number of demonstrations, set by its specificity and its reliability.** (a) A prompt read as Bayesian inference. A description states that the task w is near a value m, and the demonstrations update the posterior. The ESS of the description is the number of demonstrations it replaces. (b) The prior on w without the description (dashed) and given three descriptions (solid; sketched, not to scale), with the ESS of each when no demonstrations are in hand. Here r is the fraction of the prior variance that a relevant description leaves, p is the probability that it is relevant, and pink marks the component for an irrelevant description. The task dimension is $d = 16$, and the prior signal-to-noise ratio is $a_0 = 10$. Specificity raises the ESS (Chapter 4), and a one-in-ten chance of being irrelevant removes most of that gain (Chapter 5).

Chapter 2

Background

In this chapter, we review the ideas that the rest of the thesis builds on. Sections 2.1 to 2.3 introduce in-context learning, its interpretation as Bayesian inference, and in-context linear regression, a setting in which that interpretation can be computed exactly. Section 2.4 defines the squared error by which we compare learners and gives the error of the Bayes-optimal predictor. Sections 2.5 and 2.6 present two tools from Bayesian statistics that we later apply to descriptions: the effective sample size of a prior and the robust mixture prior. Finally, Sections 2.7 and 2.8 review the work most closely related to ours.

2.1 In-Context Learning

A language model takes a sequence of tokens, the prompt, and predicts what comes next. Brown et al. (2020) showed that large language models can perform a task specified in the prompt, without any update to their weights. This ability is called in-context learning (ICL). Brown et al. (2020) distinguished three kinds of prompts. A zero-shot prompt gives only a natural-language description of the task. One-shot and few-shot prompts add one or several *demonstrations*, solved examples of the task, before the query that the model must answer. We follow their terminology. A demonstration is one input together with its label, and we call everything else in the prompt a *description*, whether it is written as a statement about the task or as a command. A prompt can contain both, and our question is how much the description adds to the demonstrations.

How models use the demonstrations is itself an active question. Pretrained models can perform well even when the labels of the demonstrations are replaced by random ones (Min et al., 2022), which suggests that they use demonstrations partly to recognize a task they already know. Larger models, however, follow flipped in-context labels more readily (Wei et al., 2023). Lin & Lee (2024) explain both behaviours, retrieving a known task and learning a new one, within a single Bayesian model. A description raises the same question from the other side. Does a model treat it as information about the task, to be weighed against the demonstrations, or does it use it in some other way?

2.2 In-Context Learning as Bayesian Inference

Xie et al. (2022) proposed that ICL can be understood as implicit Bayesian inference. Suppose that pretraining text is generated from latent concepts. A model that predicts such text well must then infer the concept behind each document, and a prompt of demonstrations is evidence about the concept behind the prompt. In this view, a task is a latent parameter θ, and the prompt D contains demonstrations $(x_1, y_1), \ldots, (x_n, y_n)$ followed by a query x. The ideal learner predicts the query's label with the posterior predictive distribution

$$p(y \mid D) = \int p(y \mid x, \theta)\, p(\theta \mid x_{1:n}, y_{1:n})\, d\theta. \tag{2.1}$$

This view becomes testable when the prior over tasks is known. Consider a Transformer trained on prompts whose tasks are sampled from a known prior. Training on such prompts is called meta-training: the network is trained not on one task but on many, and it must infer from the context which task a prompt comes from. The posterior predictive minimizes the expected log loss over the prior (Section 2.4), and this expected log loss is exactly the training objective. A well-trained network therefore approximates the posterior predictive, whose mean is the Bayes-optimal prediction under squared error (Section 2.4) (Müller et al., 2022). For regression with a Gaussian task prior, meta-trained Transformers behave like the Bayes-optimal predictor, which is ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024), provided that they are trained on enough distinct tasks (Raventós et al., 2023). The Bayes-optimal predictor is then a reference against which a network can be measured, and we use it in this way throughout the thesis.

2.3 In-Context Linear Regression

Garg et al. (2022) introduced in-context linear regression as a controlled test of ICL. Each prompt has its own task vector $w \sim \mathcal{N}(0, I_d)$, its inputs are drawn as $x_i \sim \mathcal{N}(0, I_d)$, and its labels are $y_i = w^\top x_i$. Garg et al. (2022) found that a Transformer trained from scratch on such prompts predicts the query's label about as well as least squares. Later work added label noise, $y_i = w^\top x_i + \epsilon_i$ (Akyürek et al., 2023; Raventós et al., 2023). Raventós et al. (2023) also varied the number of distinct tasks seen in training. With few tasks, the network behaves like the Bayes-optimal predictor for a prior over those tasks alone. With enough tasks, it behaves like ridge regression, the Bayes-optimal predictor for the Gaussian prior.

With Gaussian noise and a Gaussian prior, Bayes-optimal prediction has a closed form (Gelman et al., 2013, Section 14.8). We measure labels in units of their noise, so that $y_i \sim \mathcal{N}(w^\top x_i, 1)$. Let the prior be $w \sim \mathcal{N}(\mu_0, \Sigma_0)$, and stack the inputs of n demonstrations as the rows of $X_n \in \mathbb{R}^{n \times d}$. The posterior over w is Gaussian, with covariance and mean

$$\Sigma_n = \left(\Sigma_0^{-1} + X_n^\top X_n\right)^{-1}, \qquad \mu_n = \Sigma_n\left(\Sigma_0^{-1}\mu_0 + X_n^\top y_{1:n}\right), \tag{2.2}$$

and the posterior predictive for the query is

$$y \mid D \sim \mathcal{N}\left(\mu_n^\top x,\ 1 + x^\top \Sigma_n x\right). \tag{2.3}$$

The predictive variance has two parts. The first, 1, is the label noise. The second, $x^\top \Sigma_n x$, is the remaining uncertainty about the task in the direction of the query. Under the isotropic prior $\mathcal{N}(0, a_0 I_d)$, the posterior mean $\mu_n = (X_n^\top X_n + I/a_0)^{-1} X_n^\top y_{1:n}$ is the ridge regression estimate with penalty $1/a_0$. Equation (2.2) also shows that a prior acts like additional data. Each demonstration adds $x_i x_i^\top$ to the posterior precision, and the prior adds Σ_0^{-1}. Chapter 3 uses this fact to give a description a prior of its own.

2.4 Squared Error

A predictor sees the prompt D and gives a number $\hat{y}$ for the query's label. We score it by squared error. The label is $y = w^\top x + \epsilon$, where the noise ϵ has mean 0 and

variance 1 and is independent of the prompt and the task. The expected squared error therefore splits into two parts:

$$\mathbb{E}\big[(\hat{y} - y)^2\big] = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big] + 1. \tag{2.4}$$

The second part is the noise variance, which is the same for every predictor. Following Garg et al. (2022), we therefore score a predictor by the first part alone, its *error*

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big],$$

the expected squared distance from the noiseless label $w^\top x$. An oracle that knows the task predicts $w^\top x$ and has zero error.

Given the prompt, the task is uncertain. For any prediction $\hat{y}$ computed from the prompt,

$$\mathbb{E}\big[(\hat{y} - w^\top x)^2 \mid D\big] = \big(\hat{y} - \mathbb{E}[w^\top x \mid D]\big)^2 + \mathrm{Var}(w^\top x \mid D).$$

The first term vanishes only when $\hat{y}$ is the posterior mean of $w^\top x$. When tasks are drawn from the prior, the posterior mean therefore has the smallest error of any predictor, and we call it the Bayes-optimal predictor. It is also the mean of the posterior predictive (2.1). Its error is the expected posterior variance of $w^\top x$. In linear regression, the posterior is Gaussian with covariance Σ_n (Equation (2.2)), so the posterior variance of $w^\top x$ is $x^\top \Sigma_n x$, the part of the predictive variance (2.3) that comes from uncertainty about the task. The query input is independent of the demonstrations, with $\mathbb{E}[xx^\top] = I_d$, so the error of the Bayes-optimal predictor is

$$R = \mathbb{E}\big[x^\top \Sigma_n x\big] = \mathbb{E}\,\mathrm{tr}\,\Sigma_n, \tag{2.5}$$

the expected trace of the posterior covariance. The error is zero when the posterior has concentrated on the task, and it grows with the uncertainty about the task that remains.

Two standard facts give this trace in the cases that later chapters need. With fewer demonstrations than dimensions, $n < d$, the demonstrations say nothing about $d - n$ directions, so under the prior $\mathcal{N}(0, a_0 I_d)$ the error is at least $(d - n)\, a_0$. Under a flat prior and with $n \geq d + 2$, the posterior covariance is $(X_n^\top X_n)^{-1}$, the inverse of a Wishart matrix, and the error is $\mathbb{E}\,\mathrm{tr}(X_n^\top X_n)^{-1} = d/(n - d - 1)$ (Muirhead, 1982). Chapter 4 builds its ESS of a reliable description from these two facts.

A predictor that gives a density $f(y \mid D)$ rather than a number can instead be scored by its log loss, $-\log f(y \mid D)$. The expected log loss is smallest for the true

conditional density of y given the prompt (Cover & Thomas, 2006), which is the posterior predictive when tasks are drawn from the prior. Meta-trained networks are often trained with this loss (Section 2.2), and so are the networks of Chapter 6. The mean of the posterior predictive is the posterior mean, so a network that minimizes log loss and outputs a predictive mean can be scored by squared error. We use squared error throughout for two reasons. First, it needs only the learner's answer, a number, so it applies to a language model through the number that it writes (Chapter 7). Second, the Bayes-optimal error (2.5) has the exact forms above, so the theory in Chapters 4 and 5 can work with exact errors.

Squared error also splits exactly over the components of a mixture. Suppose that a latent indicator Z selects the component that generates the task. By the law of total variance, the posterior variance is the sum of two parts:

$$\mathrm{Var}(w^\top x \mid D) = \mathbb{E}\big[\,\mathrm{Var}(w^\top x \mid D, Z) \mid D\big] + \mathrm{Var}\left(\mathbb{E}[w^\top x \mid D, Z] \mid D\right). \tag{2.6}$$

The first part is the average variance within the components. The second is the variance of the prediction across the components. Chapter 5 applies this decomposition to a description that may be irrelevant. There, the first part is the error of a predictor that is told whether the description is relevant, and the second is the price of not being told.

2.5 The Effective Sample Size of a Prior

Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information. In the conjugate normal model, this number is exact. Suppose that the observations are $y_i \sim \mathcal{N}(\theta, \sigma^2)$ and the prior is $\theta \sim \mathcal{N}(\mu_0, \sigma^2/n_0)$. After n observations with mean $\bar{y}$, the posterior is

$$\theta \mid y_{1:n} \sim \mathcal{N}\Big(\frac{n_0\mu_0 + n\bar{y}}{n_0 + n},\ \frac{\sigma^2}{n_0 + n}\Big).$$

This is the posterior that $n_0 + n$ observations would give under a flat prior if n_0 of them averaged μ_0. The prior is therefore worth n_0 observations, which is its precision measured in units of the precision of one observation. Chapter 7 uses this model, with a truncated prior. The precision c of a description's prior, defined in Chapter 3, is the same conjugate-prior ESS, counted in each direction in units of

direct observations of the task. Chapters 4 and 5 show how the ESS of a description, counted in demonstrations, differs from it.

Several definitions extend the ESS beyond the conjugate case. Morita et al. (2008) extend it to any parametric prior. Their ESS is the number of hypothetical observations that, starting from a very vague prior, would give a posterior carrying about the same information as the prior, where information is measured by the curvature of the log density. Neuenschwander et al. (2020) require an ESS to be predictively consistent: on average, the ESS of the posterior after N observations should equal the prior's ESS plus N. Other definitions let the ESS depend on the current sample, through its size or through the observed data themselves. Reimherr et al. (2021) compare a prior with a baseline prior. Their ESS is the number of observations that the baseline would need to perform as well at a given sample size, and it can be negative when the prior and the likelihood conflict. Similarly, the effective current sample size of Wiesenfarth & Calderazzo (2020) counts the observations that must be added to, or removed from, the current sample for an analysis without the prior to match the analysis with it. Finally, Reznik (2026) measures information by predictive log loss to choose the discount of a power prior. A power prior raises the likelihood of historical data, which may not match the current data, to a power between 0 and 1.

Our ESS of a description (Definition 3.1) combines elements of these definitions. Like the definitions of Reimherr et al. (2021) and Wiesenfarth & Calderazzo (2020), it depends on the number of observations already available. Like that of Reznik (2026), it measures information by predictive loss, here squared error. It counts the demonstrations that would lower the error as much as the description does.

Our ESS also differs from that of Morita et al. (2008) in what it matches. Their ESS matches information. In the conjugate normal model above, a prior of precision n_0 adds n_0 to the posterior precision whatever the number of observations, so it is worth n_0 observations at every sample size. In the same way, a description that raises the prior precision from $1/a_0$ to c in every direction is worth $(1-r)\,c$ by their measure, whatever the number of demonstrations in hand. Our ESS matches errors instead. In d dimensions, n demonstrations leave the Bayes-optimal predictor an error of $d/(n-d-1)$ under a flat prior (Section 2.4), not the d/n that their expected information would give. Matching errors therefore adds a term that grows with the dimension, the dimension term of Chapter 4. This term fades as demonstrations accumulate, and with many demonstrations in hand our ESS returns to theirs (Corollary 5.2(ii)).

2.6 Robust Mixture Priors

A prior built from historical data can conflict with the current data. Evans & Moshonov (2006) check for such prior–data conflict by asking whether the observed data are surprising under the prior predictive distribution. Clinical trials that borrow historical control data protect against conflict with a robust mixture prior (Schmidli et al., 2014). The informative prior p_I, derived from the historical data, is mixed with a vague prior p_V:

$$p(\theta) = \omega\, p_I(\theta) + (1 - \omega)\, p_V(\theta).$$

The posterior is then a mixture of the two component posteriors, and the weight on the informative component is updated in proportion to ω times its marginal likelihood. When the data agree with the informative component, its weight grows, and the trial borrows the historical information. When they conflict, the weight moves to the vague component, and the analysis falls back on the current data.

A description is a prior of the same kind. It may describe the task at hand, but it may also have been written for another task, or it may simply be wrong. Chapter 3 therefore models a description as a robust mixture prior whose weight p is the probability that the description is relevant. Chapter 5 shows how this weight affects the description's ESS.

2.7 Descriptions in Theories of ICL

A few theories of ICL include descriptions. In some, the description is a set of tokens that indicate a hypothesis class (Lin et al., 2025) or carry the mean of the inputs (Huang & Ge, 2025). In others, instructions enter a generalization bound (Tong et al., 2026). These theories generally do not measure the value of a description in demonstrations, and they do not model the possibility that it is irrelevant. For example, Huang & Ge (2025) show that a learner can use the description token to predict better, but they do not say how many demonstrations it is worth or what it is worth when it may be wrong. The Bayesian theory of Xie et al. (2022) models the prompt as demonstrations alone. We keep its view of the prompt as evidence about a latent task and add a description as a second source of that evidence, which enters through the prior. Our networks in Chapter 6 use the prefix embedding of Huang & Ge (2025), in which the description is a single token placed before the demonstrations. The

work closest to our setting is that of Zhu et al. (2026), who show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction. We measure the value of such conditioning information in demonstrations, and we allow it to be irrelevant.

2.8 Descriptions and Priors in Pretrained Language Models

Several studies measure the value of text in a prompt, or test pretrained models against a known Bayesian optimum. Cook et al. (2026) value instructions in units of training examples and find that one instruction can stand in for up to 100 execution examples in fine-tuning data. Wenliang et al. (2026) treat a prompt as conditioning for a Bayesian sequence predictor and find that an optimized prompt can be shorter than demonstrations with the same effect. Tests of pretrained models on synthetic tasks with a known optimum find that in-context evidence outweighs a stated prior (Gupta et al., 2025), that learning curves approach a Bayesian reference (Akata et al., 2026), and that models estimate a density from numbers in context like a kernel density estimator with an adaptive bandwidth (Liu et al., 2025). Arora et al. (2025) model in-context learning curves with a Bayesian scaling law that includes a task prior, and Requeima et al. (2024) show that text given alongside numbers also shapes a model's numerical predictions. We add to such a test a description with a stated specificity and a stated reliability, and we compare its ESS with the Bayes-optimal ESS. We also add a control that changes only the stated specificity, with the stated value and the demonstrations held fixed. Recall also that pretrained models can perform well with wrong demonstration labels (Min et al., 2022) and that larger models override their priors with in-context labels more readily (Wei et al., 2023). We ask the converse question: does a model weigh a description according to what the prompt says about it?

Chapter 3

Problem Setup

In this chapter, we define the setting studied in Chapters 4 to 6. It has three parts. The tasks and demonstrations are those of noisy in-context linear regression with a Gaussian task prior (Section 2.3). A description is modelled as a robust mixture prior (Section 2.6). Finally, the ESS of a description is defined through squared error (Section 2.4). Chapter 7 uses the simplest case of the same setting.

3.1 Tasks and Demonstrations

A task is a weight vector $w \in \mathbb{R}^d$. A demonstration is an input $x \sim \mathcal{N}(0, I_d)$ together with a label $y \sim \mathcal{N}(w^\top x, 1)$, so labels are measured in units of their noise. The baseline prior over tasks is $w \sim \mathcal{N}(0, a_0 I_d)$, so a_0 is the prior signal-to-noise ratio. All numerical results in Chapters 4 to 6 use $a_0 = 10$.

3.2 Descriptions

A description is a vector $m \in \mathbb{R}^d$. Its distribution depends on two parameters, r and p, which we interpret below. A latent indicator $Z \sim \text{Bernoulli}(p)$, independent of m, determines whether the description is *relevant* ($Z = 1$) or *irrelevant* ($Z = 0$):

$$
\begin{aligned}
m &\sim \mathcal{N}(0, (1 - r)\, a_0 I_d) \\
w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d) \\
w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d)
\end{aligned}
$$

In words, a relevant description says that the task lies near m, within a variance of ra_0 in every direction, while an irrelevant description is unrelated to the task. In both cases, the marginal distribution of w is the baseline prior. A relevant description splits the prior variance a_0 into two parts: the part $(1-r)\,a_0$ that m resolves and the part ra_0 that it leaves. Marginalizing over Z, the prior for w given m is the robust mixture $p\mathcal{N}(m, ra_0 I_d) + (1-p)\mathcal{N}(0, a_0 I_d)$ (Schmidli et al., 2014).

The parameter $r \in (0, 1)$ is the fraction of the prior variance that a relevant description leaves, and it sets the description's **specificity**. The smaller r is, the more specific the description. A description with r near 1 only loosely indicates the task, and one with r near 0 nearly states it. The parameter p is the description's **reliability**, and we call a description with $p = 1$ *reliable*. We write $c = 1/(ra_0)$ for the precision of a relevant description's prior, measured in demonstrations. In each direction, c is the conjugate-prior ESS of Section 2.5. We call a description *loose* when c is small compared with d, and *specific* when c exceeds d. For example, at $d = 16$ and $a_0 = 10$, the description with $r = 0.1$ ($c = 1$) is loose, and the description with $r = 0.001$ ($c = 100$) is specific. Section 4.2 also considers a reliable description that is specific in only some directions and leaves the baseline prior unchanged in the others.

3.3 Error and Effective Sample Size

A predictor sees a prompt D, which consists of the description, n demonstrations, and a query input x, and it predicts the query's label. As Garg et al. (2022) do, we score its prediction $\hat{y}$ by its error, the squared distance from the noiseless label $w^\top x$:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{3.1}$$

This is the expected squared error less the noise variance (Section 2.4). Let $R^{\text{desc}}(n)$ be a predictor's error with the description and n demonstrations, and let $R^{\text{demo}}(n)$ be its error with the n demonstrations alone. In Chapters 4 and 5, the predictor is Bayes-optimal. It predicts the posterior mean of $w^\top x$, so its error is the expected posterior variance of $w^\top x$. We write $R^{\text{desc}}_{p=1}(n)$ for the value of $R^{\text{desc}}(n)$ when the description is reliable. The ESS counts the additional demonstrations that a predictor without the description would need in order to match the predictor with it.

Definition 3.1. Given n demonstrations, the effective sample size (ESS) of a description is $\text{ESS}_n = n^* - n$, where n^* is the smallest $t \geq 0$ such that $R^{\text{demo}}(t) = R^{\text{desc}}(n)$.

Here R^{demo} is interpolated between the numbers of demonstrations at which it is known. If $R^{\text{demo}}(t) < R^{\text{desc}}(n)$ for every t, we write $\text{ESS}_n \leq -n$: the description does more harm than all n demonstrations do good.

The error without the description is known only at whole numbers of demonstrations, and for a learner other than the Bayes-optimal predictor, only at the numbers that were measured. The theory of Chapters 4 and 5 interpolates linearly between whole numbers. Chapter 7 measures the error on a sparse grid of t that doubles from one point to the next, and it interpolates in $\log t$. The grid also bounds what can be read from it. If $R^{\text{desc}}(n)$ is below R^{demo} at every t measured, up to the largest, $t_{\max}$, the crossing lies beyond the grid, and we write $\text{ESS}_n > t_{\max} - n$. If $R^{\text{desc}}(n)$ is above R^{demo} already at the smallest t measured, $t_{\min}$, we write $\text{ESS}_n \leq t_{\min} - n$.

The errors average over tasks, descriptions, and demonstrations. The ESS therefore belongs to a kind of description, determined by r and p, rather than to a particular value of m. For the Bayes-optimal predictor, the ESS is never negative. Conditioning on the description cannot raise the expected posterior variance, by the law of total variance (Section 2.4), so $R^{\text{desc}}(n) \leq R^{\text{demo}}(n)$. In addition, R^{demo} decreases in n (Section 4.3). For any other learner, the ESS is read from that learner's own two error curves (Chapters 6 and 7). It therefore measures what the description is worth to that particular learner, and it is negative when the description hurts.

Chapter 4

Specificity: Reliable Descriptions

We begin with reliable descriptions ($p = 1$) and compute their ESS for the Bayes-optimal predictor. A reliable description narrows the prior over the task, and Theorem 4.1 counts the demonstrations that would narrow it as much. Section 4.2 then turns to descriptions that inform only some directions of the task. The proofs are collected in Sections 4.3 to 4.5.

With a reliable description, the posterior over w is Gaussian. Write $W_n = X_n^\top X_n$, where the rows of X_n are the n demonstration inputs. Without the description, the posterior covariance of w is $(I/a_0 + W_n)^{-1}$ (Equation (2.2)). A reliable description replaces the baseline prior's precision $1/a_0$ with its own precision c. The Bayes-optimal prediction is the posterior mean of $w^\top x$, so its error is the expected posterior variance of $w^\top x$. The query is independent of the rest of the prompt and has $\mathbb{E}[xx^\top] = I$, so averaging over the query turns this variance into the trace of the posterior covariance (Equation (2.5)):

$$R^{\text{demo}}(n) = \mathbb{E}\,\mathrm{tr}(I/a_0 + W_n)^{-1}, \qquad R^{\text{desc}}_{p=1}(n) = \mathbb{E}\,\mathrm{tr}(cI + W_n)^{-1}. \tag{4.1}$$

4.1 The ESS of a Reliable Description

Theorem 4.1 (ESS of a reliable description)**.** *Let $p = 1$.*

(i) (Loose descriptions, worth about $(1-r)d$ demonstrations) $\mathrm{ESS}_0 \geq (1-r)\,d$*, and* $\mathrm{ESS}_0 \to (1-r)\,d$ *as $a_0 \to \infty$ with r fixed.*

(ii) (Specific descriptions, worth $c + d + 1$ demonstrations) If c is a whole number, then $\mathrm{ESS}_0 \to c + d + 1$ *as $a_0 \to \infty$ with c fixed.*

We prove the theorem in Section 4.3. The proof rests on two standard facts (Section 2.4). With $n < d$ demonstrations, the data say nothing about $d - n$ directions, so $R^{\text{demo}}(n) \geq (d - n)\, a_0$. With a flat baseline prior and $n \geq d + 2$ demonstrations, $R^{\text{demo}}(n) = d/(n - d - 1)$, the trace of the mean of an inverse Wishart matrix (Muirhead, 1982). A reliable description with no demonstrations leaves the error $R^{\text{desc}}_{p=1}(0) = d/c = ra_0 d$. The ESS is the number of demonstrations at which these curves fall to this value.

The ESS has two parts. The first, which we call the *dimension term*, counts dimensions. At a high signal-to-noise ratio, each of the first d demonstrations removes the uncertainty in one direction. A loose description removes a fraction $1 - r$ of the prior variance in every direction, so it is worth about $(1 - r)d$ of these demonstrations. A specific description removes nearly all of the variance, so it is worth about d.

The second part applies to specific descriptions, which are also worth c demonstrations. This is the ESS that Morita et al. (2008) give a normal prior, the ratio of the observation variance to the prior variance (Section 2.5). Their ESS matches information. A description that raises the prior precision from $1/a_0$ to c is therefore worth $c - 1/a_0 = (1 - r)\, c$ demonstrations under their definition, at every n. Our ESS matches errors instead. With a flat prior, n demonstrations leave the error $d/(n-d-1)$, not the error d/n that their expected information would suggest. Setting $d/(n-d-1)$ equal to the description's error d/c gives $n = c+d+1$, while setting d/n equal to it gives $n = c$. The dimension term is the gap between the two, which is $d+1$ for a specific description. It fades as demonstrations accumulate (Corollary 5.2(ii) in Chapter 5).

At $d = 16$ and $a_0 = 10$, a description that leaves a tenth of the prior variance ($r = 0.1$, $c = 1$) is worth 15.6 demonstrations, compared with $(1 - r)d = 14.4$. A description that leaves a thousandth ($r = 0.001$, $c = 100$) is worth 117, which matches $c + d + 1$ (Figure 4.1). With many demonstrations, the dimension term fades, because the demonstrations have already removed that uncertainty. At $r = 0.1$, for example, the ESS falls from 15.6 with no demonstrations to 1.1 with 100.

4.2 Descriptions of Part of the Task

A description can be specific about some aspects of a task and silent about others. We model such a description as a reliable description that leaves the fraction r of the

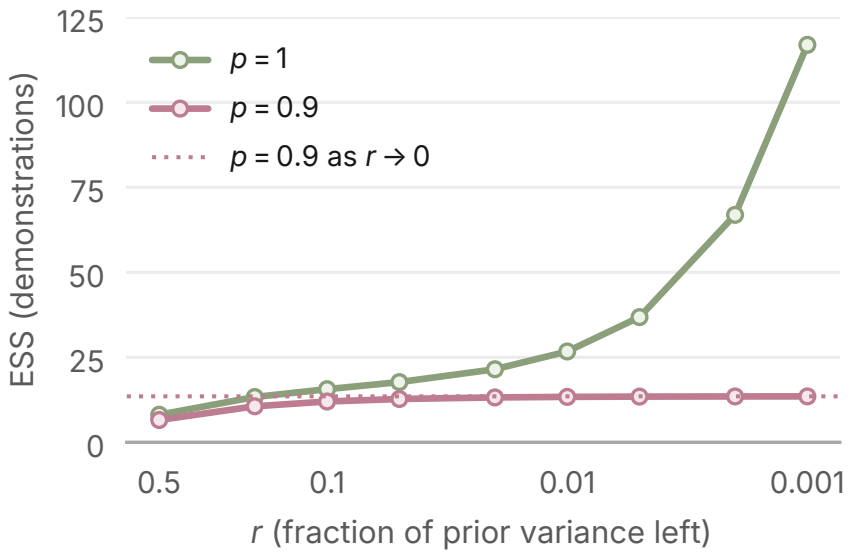


Figure 4.1: **Specificity raises the ESS of a reliable description without bound, but the ESS of a description that may be irrelevant stays below a bound.** The Bayes-optimal ESS of a description with no demonstrations in hand is plotted against r, the fraction of the prior variance that the description leaves. The two curves show a reliable description ($p = 1$) and a description that is relevant with probability 0.9. The dotted line marks the limit of the $p = 0.9$ ESS as $r \to 0$. Here $d = 16$ and $a_0 = 10$. Each point averages eight batches of 4,000 prompts, and the standard errors are below 0.02 demonstrations.

prior variance in $k < d$ directions and all of the prior variance in the other $d - k$. Its prior precision is then c in those k directions and $1/a_0$ in the rest.

Proposition 4.2 (ESS of a partial description)**.** *Let $p = 1$ and $k < d$.*

(i) (Without demonstrations, worth about k demonstrations) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1 - r)\, k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.

(ii) (With many demonstrations, worth $(1 - r)ck/d$ demonstrations) As $n \to \infty$, $\mathrm{ESS}_n \to (1 - r)\, c\, k/d$.

We prove the proposition in Section 4.5. Without demonstrations, a partial description is worth the number of directions that it pins down, and its precision adds nothing. The reason is that the error left in the directions it is silent about, $(d - k)\, a_0$, dwarfs the error that its precision removes. The value $c + d + 1$ of Theorem 4.1(ii) therefore needs a description of every direction.

Table 4.1: **A specific partial description gains value as demonstrations accumulate.** The table gives the Bayes-optimal ESS of a reliable description of k of the $d = 16$ directions, with $a_0 = 10$ and n demonstrations in hand. The row $k = 16$ is a complete description. The last column of each block is the limit $(1 - r)ck/d$ of Proposition 4.2(ii). Each value averages eight batches of 2,000 prompts, and the standard errors are at most 0.23.

	loose, $r = 0.1$ ($c = 1$)					specific, $r = 0.001$ ($c = 100$)				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

Once demonstrations have resolved the other directions, the description is worth the precision that it adds, $k(1 - r)c$, measured in units of the information d that one demonstration carries. When this value exceeds k, which happens about when $(1 - r)c > d$, a partial description gains value as demonstrations accumulate. Table 4.1 shows this at $d = 16$, $a_0 = 10$, and $r = 0.001$. A description of half of the directions is worth 8.1 demonstrations with none in hand and 36 with 100 in hand. Over the same range, a complete description falls from 117 to 109. Demonstrations thus substitute for a complete description and complement a partial one. A loose description ($r = 0.1$) loses value as demonstrations accumulate, whether it is partial or complete.

4.3 Proof of Theorem 4.1

The proof compares the error curve without the description with the error that a reliable description leaves. We first fix notation and collect three standard facts about Wishart matrices.

Notation and standard facts. Throughout the proofs, $\varepsilon = 1/a_0$, the matrix $X_n \in \mathbb{R}^{n \times d}$ holds the demonstration inputs as rows, and $W_n = X_n^\top X_n$. For $\kappa > 0$, let

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I + W_n)^{-1}$$

be the error with n demonstrations under a Gaussian prior of precision κ in every direction. By (4.1), $R^{\mathrm{demo}} = R^{\varepsilon}$ and $R^{\mathrm{desc}}_{p=1} = R^{c}$. We use three facts.

(W1) For $n < d$, W_n has rank n almost surely, so $(\kappa I + W_n)^{-1}$ has the eigenvalue $1/\kappa$ with multiplicity $d - n$.

(W2) For $n \geq d$ and a fixed unit vector e, $1/(e^\top W_n^{-1} e) \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence $\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$, and $\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = \infty$ for $n \in \{d, d+1\}$.

(W3) As $n \to \infty$ with d fixed, $\mathbb{E}\,\mathrm{tr}\, W_n^{-2} = d/n^2 + O(n^{-3})$ and $\mathbb{E}\,\mathrm{tr}\, W_n^{-3} = O(n^{-3})$ (von Rosen, 1988).

By the Sherman–Morrison formula, each new demonstration lowers $(\kappa I + W_n)^{-1}$ by a positive semidefinite matrix that is nonzero almost surely. So R^{κ} is strictly decreasing in n. Since $R^{\kappa}(n) \leq \mathbb{E}\,\mathrm{tr}\, W_n^{-1}$, (W2) shows that it tends to 0.

Proof of Theorem 4.1. Without demonstrations, the error that the description leaves is the trace of its prior covariance I/c, so $R^{\mathrm{desc}}_{p=1}(0) = d/c = r a_0 d$.

(i). For $n < d$, (W1) gives $R^{\varepsilon}(n) \geq (d - n)\, a_0$, because the other eigenvalues are positive. For $n \geq d$, the bound holds trivially, because its right side is not positive. The bound is linear in n, so the interpolated R^{demo} satisfies $R^{\mathrm{demo}}(t) \geq (d - t)\, a_0$ at every real $t \geq 0$. For $t < (1 - r)d$, the right side exceeds $r a_0 d$, so $\mathrm{ESS}_0 \geq (1 - r)d$. Now let $a_0 \to \infty$ with r fixed. Then $R^{\varepsilon}(n)/a_0 = \mathbb{E}\,\mathrm{tr}(I + a_0 W_n)^{-1}$. Each eigenvalue $1/(1 + a_0\lambda)$ of this matrix lies in $(0, 1]$, and it tends to 1 if $\lambda = 0$ and to 0 otherwise. By (W1), and because W_n has full rank almost surely for $n \geq d$, exactly $(d - n)^+$ eigenvalues of W_n are zero. By dominated convergence, $R^{\varepsilon}(n)/a_0 \to (d - n)^+$. So the interpolated $R^{\mathrm{demo}}(t)/a_0$ tends to $(d - t)^+$ at every t. This limit falls through the description's error over a_0, rd, only at $t = (1 - r)d$. For any $0 < \delta \leq rd$, the limit at $t = (1 - r)d + \delta$ is $rd - \delta$, so $R^{\mathrm{demo}}(t) < r a_0 d$ there once a_0 is large. As R^{demo} is decreasing, $\mathrm{ESS}_0 < (1 - r)d + \delta$ for such a_0, and with the lower bound, $\mathrm{ESS}_0 \to (1 - r)d$.

(ii). With c fixed, $R^{\mathrm{desc}}_{p=1}(0) = d/c$ does not depend on a_0. In contrast, $R^{\varepsilon}(n)$ increases as a_0 grows. By monotone convergence, it tends to $+\infty$ for $n \leq d + 1$, by (W1) for $n < d$ and by (W2) for $n \in \{d, d + 1\}$. For $n \geq d + 2$, it tends to $\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = d/(n - d - 1)$. So the interpolated curve tends to $+\infty$ below $d + 2$ and to the linear interpolation of $d/(n - d - 1)$ above it. This limit is strictly decreasing, and for whole c it equals d/c at $n = c + d + 1$. The root that defines ESS_0 therefore

increases to $c + d + 1$. □

4.4 Extra Precision with Many Demonstrations

The proofs of Proposition 4.2(ii) and of Corollary 5.2(ii) in Chapter 5 need the error to second order in $1/n$, which we derive here. Since $\kappa I + W_n$ and W_n commute,

$$(\kappa I + W_n)^{-1} = W_n^{-1} - \kappa\, W_n^{-2} + \kappa^2\, W_n^{-2}(\kappa I + W_n)^{-1}.$$

Because $(\kappa I + W_n)^{-1} \preceq W_n^{-1}$, the last term lies between 0 and $\kappa^2 W_n^{-3}$. Taking expected traces with (W2) and (W3) gives, for fixed d and κ,

$$R^{\kappa}(n) = \frac{d}{n - d - 1} - \frac{\kappa\, d}{n^2} + O(n^{-3}). \tag{4.2}$$

Each further demonstration lowers the first term by about d/n^2. A prior of precision κ lowers the error by κ times as much. Once n is large, such a prior is therefore worth κ demonstrations.

4.5 Proof of Proposition 4.2

Let $\Lambda = \mathrm{diag}(cI_k, \varepsilon I_{d-k})$ be the prior precision that the partial description gives. The error with the description is then $\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1}$. The law of W_n is invariant under rotations, so it does not matter which k directions the description informs.

Proof of Proposition 4.2. (i). Without demonstrations, the error with the description is $\mathrm{tr}\,\Lambda^{-1} = k/c + (d-k)\,a_0 = (kr + d - k)\,a_0$. As in the proof of Theorem 4.1(i), $R^{\mathrm{demo}}(t) \geq (d-t)\,a_0$ at every $t \geq 0$, so $\mathrm{ESS}_0 \geq (1-r)k$. Also as there, $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$, and the convergence is uniform in t because the curves are monotone and the limit is continuous. With r fixed, the error with the description over a_0 is $d - (1-r)k$. With c fixed, it is $d - k + k/(c\,a_0)$, which tends to $d - k$. As $k < d$, the limit curve passes through each of these values at a single point, $(1-r)k$ or k, with slope -1. The root that defines ESS_0 therefore converges to that point.

(ii). Write $A = \Lambda + W_n$. Since $A^{-1} = W_n^{-1} - W_n^{-1}\Lambda A^{-1}$, expanding twice gives

$$A^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda A^{-1}.$$

By rotation invariance, $\mathbb{E}W_n^{-2} = (\mathbb{E}\,\mathrm{tr}\,W_n^{-2}/d)\,I$. By (W3), the middle term therefore has expected trace $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$. For the last term, we use $|\,\mathrm{tr}(MB)| \le \|M\|\,\mathrm{tr}\,B$ for $B \succeq 0$, with $M = W_n^{-1}\Lambda W_n^{-1}\Lambda$ and $B = A^{-1} \preceq W_n^{-1}$. The trace of the last term is then at most $\|\Lambda\|^2\|W_n^{-1}\|^2\,\mathrm{tr}\,W_n^{-1} \le d\,\|\Lambda\|^2\,\mathrm{tr}\,W_n^{-3}$, whose expectation is $O(n^{-3})$ by (W3). Hence

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n-d-1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}),$$

which is (4.2) with κd replaced by $\mathrm{tr}\,\Lambda$. With (4.2) for $R^{\mathrm{demo}} = R^{\varepsilon}$, uniformly over integers $0 \le t \le T$ for any fixed T,

$$R^{\mathrm{demo}}(n+t) - \mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n^2}\Big[\frac{\mathrm{tr}\,\Lambda - d\varepsilon}{d} - t + O(1/n)\Big].$$

The argument in the proof of Corollary 5.2(ii) with $p = 1$ (Section 5.4) then gives $\mathrm{ESS}_n \to (\mathrm{tr}\,\Lambda - d\varepsilon)/d = k(c-\varepsilon)/d = (1-r)\,c\,k/d$. □

The same steps give the limit $(\mathrm{tr}\,\Lambda - d\varepsilon)/d$ for a reliable description with any Gaussian prior precision Λ. With many demonstrations in hand, a description is therefore worth the precision that it adds, averaged over directions.

Chapter 5

Reliability: Descriptions That May Be Irrelevant

We now turn to descriptions that may be irrelevant ($p < 1$). The Bayes-optimal prediction then mixes two predictions. The first, $\mu_1^\top x$, takes the description as relevant, and the second, $\mu_0^\top x$, ignores it. Here μ_1 and μ_0 are the posterior means of w under the two readings of the description. The two predictions are weighted by the posterior probability π_n that the description is relevant (Section 2.6). Proposition 5.1 gives the price of not knowing whether the description is relevant, and it gives the error in closed form when no demonstrations are in hand. Corollary 5.2 turns the proposition into bounds on the ESS. Finally, Section 5.3 shows what happens to a predictor that does not hedge but trusts every description.

5.1 The Price of Not Knowing Relevance

Proposition 5.1 (Error of a description that may be irrelevant)**.** *For every $n \geq 0$,*

$$R^{\text{desc}}(n) \;=\; p\, R^{\text{desc}}_{p=1}(n) + (1-p)\, R^{\text{demo}}(n) + \mathcal{V}_n, \tag{5.1}$$

where $\mathcal{V}_n = \mathbb{E}\big[\pi_n(1-\pi_n)\big((\mu_1-\mu_0)^\top x\big)^2\big] \geq 0$. With no demonstrations in hand,

$$R^{\text{desc}}(0) = \big(1 - p^2(1-r)\big)\, a_0 d. \tag{5.2}$$

We prove the proposition in Section 5.4. The decomposition (5.1) is the law of total variance over Z (Equation (2.6)). Its first two terms are the error of a *cheating*

predictor, one that is told whether the description is relevant. With probability p, this predictor holds a reliable description, and with probability $1-p$, it holds only the demonstrations. To the cheating predictor, a description that may be irrelevant is worth as much as a reliable one when it is relevant, and nothing when it is not.

The last term, $\mathcal{V}_n$, is the price of not being told. It is the variance of the prediction between the two readings of the description. The price is small when the two readings agree. It is also small when the demonstrations make one reading far likelier than the other, because $\pi_n(1-\pi_n)$ is then near zero. With no demonstrations, the price grows as a more specific description pulls the two readings apart. With demonstrations, however, a specific description is also easier to tell apart from an irrelevant one.

With no demonstrations, Equation (5.2) puts the two predictors side by side. The baseline prior leaves the error $a_0 d$. The cheating predictor removes a fraction $p(1-r)$ of this error, and the Bayes-optimal predictor removes a fraction $p^2(1-r)$. Not being told therefore costs $p(1-p)(1-r)\,a_0 d$.

5.2 The ESS of a Description That May Be Irrelevant

Corollary 5.2 (ESS of a description that may be irrelevant)**.**

(i) (Without demonstrations) If $p < 1$, then $\mathrm{ESS}_0 \le n_p^{\max}$, where $n_p^{\max}$ solves $R^{\mathrm{demo}}(n_p^{\max}) = (1-p)\,R^{\mathrm{demo}}(0)$ for the interpolated R^{demo}. This bound depends on d, a_0, and p but not on r. Also $\mathrm{ESS}_0 \ge p^2(1-r)\,d$, and as $a_0 \to \infty$ with r fixed, $n_p^{\max} \to p\,d$ and $\mathrm{ESS}_0 \to p^2(1-r)\,d$.

(ii) (With many demonstrations) As $n \to \infty$, $\mathrm{ESS}_n \to (1-r)\,c$ if $p = 1$, and if $p < 1$,

$$\limsup_{n\to\infty} \mathrm{ESS}_n \le p\,(1-r)\,c.$$

Without demonstrations, the ESS is bounded. Refining a description from $r = 0.1$ to $r = 0.001$ raises its ESS from 15.6 to 117 if the description is reliable, but only from 12.0 to 13.5 if it is relevant with probability 0.9. The reason is a floor on the error. However specific the description, the average error cannot fall below the contribution of irrelevant descriptions, $(1-p)R^{\mathrm{demo}}(0)$. Corollary 5.2(i) turns this floor into a bound on the ESS: $n_p^{\max} = 15.6$ demonstrations at $d = 16$, $a_0 = 10$, and $p = 0.9$. This bound is the ESS of a reliable description that leaves a fraction $1-p$ of the prior variance, because such a description leaves the same error, $(1-p)\,a_0 d$. The

bound holds even for the cheating predictor, which is told whether the description is relevant. It therefore comes from irrelevant descriptions being worth nothing, not from uncertainty about relevance. That uncertainty lowers the ESS further. At a high signal-to-noise ratio, it lowers the ESS from about pd to about p^2d. At $a_0 = 10$, as $r \to 0$, the ESS approaches 13.5 (the dotted line in Figure 4.1), compared with $p^2d = 13.0$.

With many demonstrations, the ESS is at most p times that of a reliable description. Demonstrations shrink the error in the irrelevant case, $R^{\text{demo}}(n)$, so the floor $(1-p)R^{\text{demo}}(n)$ falls and the ESS can grow. In the limit, a reliable description is worth $(1-r)\,c$ demonstrations. The dimension term has faded, and what remains is the precision that the description adds to the baseline prior. A description that may be irrelevant is then worth at most p times as much. With 100 demonstrations in hand, the $r = 0.001$ description is worth 109 demonstrations if it is reliable and 86.6 if $p = 0.9$, compared with limits of 99.9 and at most 89.9. The reliable value exceeds its limit because at $n = 100$ the dimension term has not yet faded completely.

5.3 Full Trust Can Be Worse Than No Description

The Bayes-optimal predictor hedges between the two readings of the description. A *fully trusting* predictor does not: it treats every description as relevant and predicts as if $p = 1$. With no demonstrations, its error is

$$R^{\text{trust}}(0) = \big(1 - (2p-1)(1-r)\big)\, a_0 d \tag{5.3}$$

(derived in Section 5.5). The fully trusting predictor removes a fraction $(2p-1)(1-r)$ of the prior error, compared with $p^2(1-r)$ for the Bayes-optimal predictor. With no demonstrations, it does worse than ignoring the description only when $p < 1/2$. Demonstrations do not rescue it. They show whether a specific description is relevant, but the fully trusting predictor keeps predicting near m. As $r \to 0$, its error tends to $2(1-p)\,a_0 d$ at every n, while the error without the description tends to 0 as n grows. At $d = 16$, $a_0 = 10$, and $p = 0.9$, the fully trusting predictor with the $r = 0.001$ description is worse than no description from 15 demonstrations on.

5.4 Proofs of Proposition 5.1 and Corollary 5.2

Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ be the prompt, including the query input, and let $\pi_n = P(Z = 1 \mid D_n)$ be the posterior probability that the description is relevant.

Step 1: the two readings. Given D_n and $Z = 1$, w is Gaussian with the posterior mean μ_1 and covariance Σ_1 of a reliable description, that is, the posterior (2.2) under the prior $\mathcal{N}(m, ra_0 I)$. Given D_n and $Z = 0$, the description m is independent of w. So w is Gaussian with the posterior mean μ_0 and covariance Σ_0 from the demonstrations alone. At $n = 0$, $\pi_0 = p$. Observing m says nothing about Z, because m has the same distribution $\mathcal{N}(0, (1-r)a_0 I)$ in both cases, and x is independent of Z.

Step 2: the decomposition. The Bayes-optimal prediction is $\mathbb{E}[w^\top x \mid D_n] = (\pi_n \mu_1 + (1-\pi_n)\mu_0)^\top x$, and its error is $\mathbb{E}\,\mathrm{Var}(w^\top x \mid D_n)$. By the law of total variance over Z (2.6),

$$\mathrm{Var}(w^\top x \mid D_n) = \pi_n\, x^\top \Sigma_1 x + (1-\pi_n)\, x^\top \Sigma_0 x + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2.$$

The first two terms are the expected conditional variance, and the last is the variance of the conditional mean. For any function g of the prompt, $\mathbb{E}[\pi_n\, g(D_n)] = \mathbb{E}[Z\, g(D_n)] = p\,\mathbb{E}[g(D_n) \mid Z = 1]$. Given $Z = 1$, the prompt has the distribution of a prompt with a reliable description, so $\mathbb{E}[\pi_n\, x^\top \Sigma_1 x] = p\, R^{\mathrm{desc}}_{p=1}(n)$. In the same way, $\mathbb{E}[(1-\pi_n)\, x^\top \Sigma_0 x] = (1-p)\, R^{\mathrm{demo}}(n)$. Taking expectations of the three terms gives (5.1).

Step 3: no demonstrations. At $n = 0$, $\pi_0 = p$, $\mu_1 = m$, and $\mu_0 = 0$, so the Bayes-optimal prediction is $p\, m^\top x$. In each coordinate, $\mathrm{Var}(w_i) = a_0$ and $\mathrm{Var}(m_i) = (1-r)\, a_0$. When the description is relevant, w_i is m_i plus independent noise, and when it is irrelevant, w_i is independent of m_i. Hence $\mathrm{Cov}(w_i, m_i) = p\,(1-r)\, a_0$, and

$$\mathbb{E}(w_i - p\, m_i)^2 = a_0 - 2p^2(1-r)\, a_0 + p^2(1-r)\, a_0 = \big(1 - p^2(1-r)\big)\, a_0.$$

The query is independent of (w, m) with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}\|w - p\, m\|^2$ is d times this value, which is (5.2). This completes the proof of Proposition 5.1.

Proof of Corollary 5.2. (i). At $n = 0$, (5.1) with $\mathcal{V}_0 \geq 0$ gives $R^{\mathrm{desc}}(0) \geq (1-p)\, R^{\mathrm{demo}}(0)$ for every r, because $R^{\mathrm{desc}}_{p=1}(0) \geq 0$. Since R^{demo} is strictly decreasing and tends to 0 (Section 4.3), there is a finite $n_p^{\max}$ with $R^{\mathrm{demo}}(n_p^{\max}) = (1-p)\, R^{\mathrm{demo}}(0)$, and

the root that defines ESS_0 cannot exceed it. By (5.2), $R^{\mathrm{desc}}(0) = \big(d - p^2(1-r)\,d\big)\,a_0$, and since $R^{\mathrm{demo}}(0) = a_0 d$, $(1-p)\,R^{\mathrm{demo}}(0) = (d-pd)\,a_0$. The proof of Theorem 4.1(i) gives the bound $R^{\mathrm{demo}}(t) \geq (d-t)\,a_0$ and the limit $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$. The bound gives $\mathrm{ESS}_0 \geq p^2(1-r)\,d$, and the limit gives $\mathrm{ESS}_0 \to p^2(1-r)\,d$ and $n_p^{\max} \to pd$.

(ii). We first sketch the idea. With many demonstrations, each further demonstration lowers the error by about d/n^2, and a prior of precision κ lowers it by κ times as much (Section 4.4). A reliable description raises the precision from ε to c, so it saves about $c - \varepsilon = (1-r)c$ demonstrations. A description that is relevant with probability p saves about p times as much, and the price $\mathcal{V}_n$ can only lower its value further.

Formally, fix d, a_0, and r, and read every O below as $n \to \infty$ with these quantities fixed. A reliable description leaves the prior $\mathcal{N}(m, ra_0 I)$, which has precision c in every direction, and its error does not depend on m. Hence $R^{\mathrm{desc}}_{p=1} = R^c$ in the notation of Section 4.3, while $R^{\mathrm{demo}} = R^\varepsilon$. Let $L(n) = p\,R^c(n) + (1-p)\,R^\varepsilon(n)$ be the first two terms of (5.1). By (4.2), uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\Big[p\,(c-\varepsilon) - t + O(1/n)\Big].$$

The bracket is linear in t, so the same expansion holds for the interpolated R^{demo} at real $t \in [0, T]$. The interpolated curve therefore equals $L(n)$ at a unique point $n + \eta_n$, with $\eta_n = p\,(c-\varepsilon) + O(1/n)$, where $c - \varepsilon = (1-r)\,c$. If $p = 1$, then $R^{\mathrm{desc}} = L$, and $\mathrm{ESS}_n = \eta_n \to (1-r)\,c$. If $p < 1$, then $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$. Since R^{demo} is decreasing, $\mathrm{ESS}_n \leq \eta_n$, and hence $\limsup_n \mathrm{ESS}_n \leq p\,(1-r)\,c$. □

5.5 Derivation of Equation (5.3)

The idea is that, when the description is irrelevant, the fully trusting predictor centres its prediction on an unrelated mean. With no demonstrations, it predicts $m^\top x$. With the moments of Step 3 in Section 5.4,

$$\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1-r)\,a_0 + (1-r)\,a_0 = \big(1 - (2p-1)(1-r)\big)\,a_0,$$

and summing over the d coordinates gives (5.3). This error exceeds $R^{\mathrm{demo}}(0) = a_0 d$ exactly when $p < 1/2$.

With n demonstrations, the fully trusting predictor predicts $\mu_1^\top x$ whatever the value of Z, where

$$\mu_1 = m + (cI + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

As $r \to 0$ with a_0 and n fixed, $c \to \infty$ and $\mu_1 \to m$ in mean square, so the error tends to $\mathbb{E}\|w - m\|^2$. This limit is $r a_0 d \to 0$ for a relevant description and $(2 - r)\, a_0 d \to 2 a_0 d$ for an irrelevant one, which gives $2(1 - p)\, a_0 d$ in all. Since $R^{\mathrm{demo}}(n) \to 0$, for every $p < 1$ there is an n from which full trust does worse than no description once r is small enough.

Chapter 6

Meta-Trained Transformers

The Bayes-optimal predictor of Chapters 4 and 5 knows how specific and how reliable descriptions are. A network must instead learn both properties from its training data. In this chapter, we meta-train Transformers (Section 2.2) on the setting of Chapter 3 and ask two questions. How close do the networks come to the Bayes-optimal ESS? And what sets how much they trust a description?

6.1 Setup

Networks. We encode prompts with the prefix embedding of Huang & Ge (2025). A prompt consists of an optional description token that carries m, up to 20 demonstration tokens, and a query token. The network is a Transformer (Vaswani et al., 2017) of the same size as that of Garg et al. (2022) (12 layers, 8 heads, and width 256) and is trained with their optimizer. Unlike theirs, it uses bidirectional attention and no positional encoding (Section 6.3). To keep training short, we use $d = 5$, with $a_0 = 10$ as before. The network outputs a mean and a log variance for the query's label and is trained with log loss, as in Zhu et al. (2026). The mean that minimizes this loss is the posterior mean, which is the Bayes-optimal prediction under squared error. We therefore score the network's mean by its squared error (3.1), the same error that we use for the Bayes-optimal predictor. The description token carries only m, and half of the training prompts omit it, so each network must learn r and p from its training distribution. We train a network for each combination of specificity ($r = 0.1$, loose; $r = 0.01$, specific) and reliability ($p \in \{1, 0.99, 0.9\}$). For each specific setting, we

train three networks that differ only in their random seed. For each loose setting, we train two. Section 6.3 describes four further networks, at $r = 0.5$ and $r = 0.05$.

Evaluation. We test each network on 20,000 fresh prompts at every n from 0 to 20, with and without a description. Networks that differ only in their seed are tested on the same prompts. Each network is tested at its training reliability and at $p = 1$ and $p = 0.9$. The networks trained with the second and third seeds are also tested at $p = 0.99$ (Table 6.1). We read each network's ESS from its own error curves, following Definition 3.1. Without a description, the networks' errors are close to the Bayes-optimal predictor's. At $n = 0$, where the Bayes-optimal error is about 50, every network's error is within 0.21 of it. Averaged over the numbers of demonstrations on which a network was trained, the difference is at most 0.3.

We compare each network's ESS with that of two Bayes-optimal predictors. The *trained* Bayes-optimal predictor uses the reliability on which the network was trained. The *calibrated* Bayes-optimal predictor uses the reliability of the test prompts. Finally, to summarize how much a network trusts a description, we define its *implied trust*. With no demonstrations, the Bayes-optimal prediction at reliability q is $q\, m^\top x$. A network's implied trust is the value of q for which this prediction is closest to the network's prediction in squared error. It equals the regression coefficient of the network's prediction on $m^\top x$, and it can exceed 1 (Table 6.1).

6.2 Results

Networks come close to the Bayes-optimal ESS. We first train and test each network at the same reliability (Figure 6.1a). For the specific description at $p = 1$, the network's ESS is 14.3 ± 0.5 (mean $\pm$ standard deviation over three seeds). On the same prompts, the Bayes-optimal ESS is 16.1. On 200,000 prompts, it is 15.9, close to the value $c + d + 1 = 16$ of Theorem 4.1(ii), since $c = 1/(ra_0) = 10$. At $p = 0.99$, the network's ESS is 9.3 ± 0.2, compared with 9.6 for the Bayes-optimal predictor on the same prompts. At $p = 0.9$, it is 4.66 ± 0.01, compared with 4.63. Because the three networks share their test prompts, these standard deviations reflect training alone and leave out the variation from prompts. We therefore compare each network with the Bayes-optimal predictor on the same prompts. Table 6.1 instead gives the Bayes-optimal ESS on 200,000 prompts. At $p = 0.99$, that value (8.96) lies below the

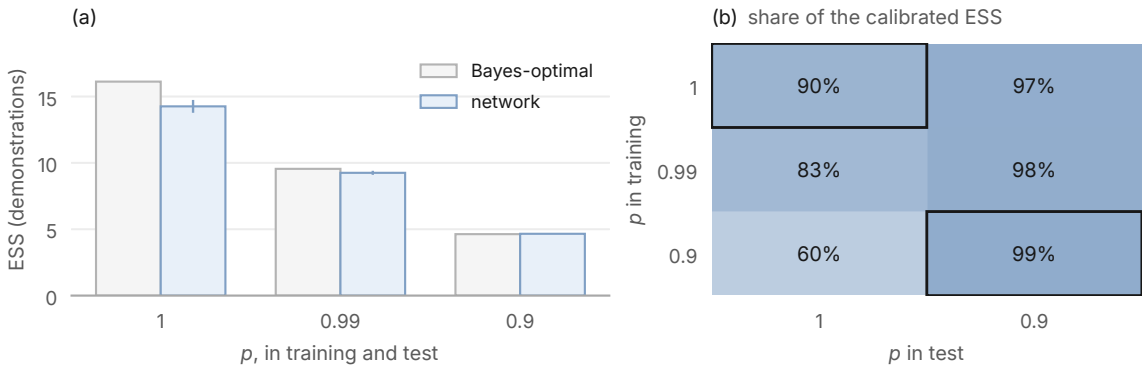


Figure 6.1: **Networks come close to the Bayes-optimal ESS and take their trust in a description from training.** All panels use the specific description ($r = 0.01$), $d = 5$, $a_0 = 10$, and no demonstrations. Each network ESS shown is the mean over three networks trained with different random seeds, and the error bars show ± one standard deviation across them. (a) Networks trained and tested at the same reliability p. The bars show the Bayes-optimal ESS on the same prompts and the network's ESS, which is read from the network's own no-description curve. (b) Networks trained at one reliability and tested at another. Each cell shows the network's ESS as a share of the calibrated predictor's, and outlined cells mark cases where the training and test reliabilities match. Table 6.1 gives every value, including those for the loose description ($r = 0.1$).

networks' mean ESS, whereas the Bayes-optimal ESS on the networks' own prompts (9.6) lies above it. A network's ESS can exceed the Bayes-optimal ESS on the same prompts, as it does at $p = 0.9$, because it is read from the network's own no-description curve rather than from the Bayes-optimal one. For the loose description, the network's ESS is within 0.1 of the Bayes-optimal ESS at every reliability. At $p = 1$, for example, it is 5.85 ± 0.03, compared with 5.89 (Table 6.1).

ESS as demonstrations accumulate. Figure 6.2 shows each network's ESS_n at its training reliability as demonstrations accumulate. The test prompts hold at most 20 demonstrations, so the largest ESS_n that can be read is $20 - n$. For the loose description, the networks' ESS_n follows the Bayes-optimal ESS_n to within 0.6 at every n the grid can show. For the specific description, the networks fall short of the Bayes-optimal ESS_n by up to 2.4 demonstrations at $p = 1$, 2.9 at $p = 0.99$, and 1.4 at

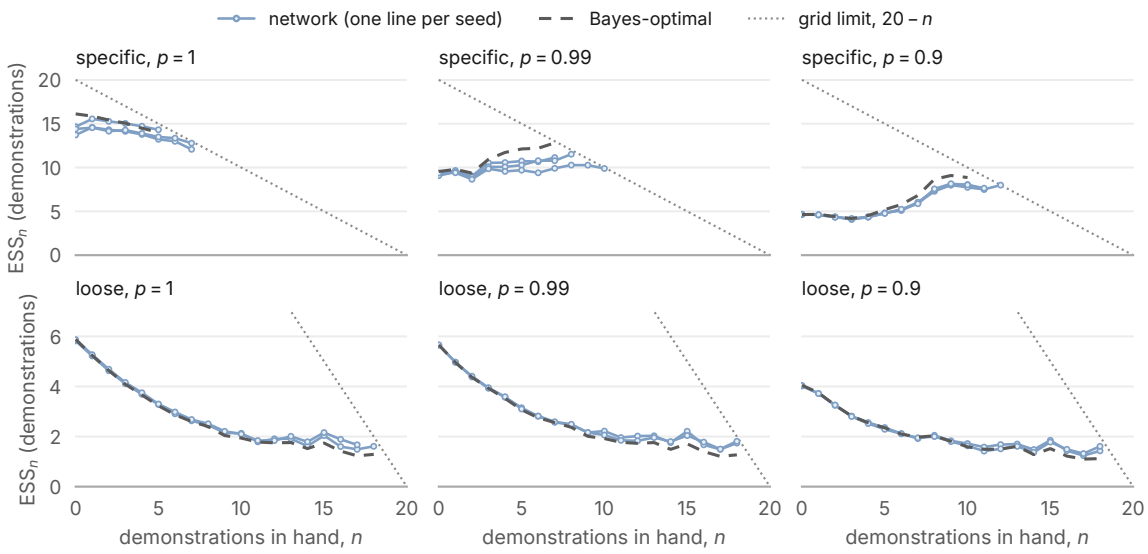


Figure 6.2: **The networks' ESS as demonstrations accumulate.** Each panel shows ESS_n for the meta-trained networks, with one line per training seed, and for the Bayes-optimal predictor on the same prompts. The networks are trained and tested at the same reliability p. The top row uses the specific description ($r = 0.01$), and the bottom row uses the loose description ($r = 0.1$). Here $d = 5$ and $a_0 = 10$, and each network is tested on 20,000 prompts. A curve stops where the error without the description would reach the error with it only after more than the 20 demonstrations of the grid. The dotted line, $20 - n$, is the largest ESS that the grid can show.

$p = 0.9$. They exceed it by at most 0.3. Both comparisons hold until the ESS can no longer be read within the grid. With five demonstrations in hand, the three networks trained at $p = 0.9$ have an ESS of 4.83, 4.79, and 4.77, compared with 5.20 for the Bayes-optimal predictor.

Training sets a network's trust. With no demonstrations, the prompt carries no information about relevance (Section 5.4, Step 1). Any learner must therefore take its trust in a description from its training. The networks show how much of that trust they take (Figure 6.1b). The networks trained at $p = 0.9$ keep hedging even when every test description is relevant. Their implied trust is 0.87 to 0.89 across the three seeds. Their ESS on reliable descriptions is 9.5 ± 0.4, whereas the calibrated predictor reaches 15.9 and the trained one 10.8.

The networks trained on specific descriptions at $p = 1$ or $p = 0.99$ treat every description as relevant when no demonstrations are shown. Their implied trust lies between 0.95 and 1.02, so the two training reliabilities cannot be told apart without demonstrations. Demonstrations separate them. On prompts in which one description in ten is irrelevant, the description lowers the error of the networks trained at $p = 0.99$ at every n, as it does for the Bayes-optimal predictor. It lowers their error by 3.5 to 3.7 with six demonstrations and by 0.08 to 0.13 with 20. The networks trained at $p = 1$ instead trust the description fully. As Section 5.3 predicts, full trust costs little with no demonstrations. Their ESS is then 4.58 ± 0.03, compared with 4.70 for the calibrated predictor. From six demonstrations on, however, their error is higher with the description than without it, in all three seeds. The fully trusting Bayes-optimal predictor behaves in the same way. With eight demonstrations, for example, the first of these networks has an error of 4.6 with the description and 2.3 without it.

6.3 Training Details

Each demonstration token holds an input together with its label, and two indicator coordinates mark the token type. There is no positional encoding. The Transformer has 12 layers, 8 heads, and width 256. It is trained with Adam (learning rate 10^{-4}) for 40,000 steps at batch size 1,024, on fresh prompts at every step and without a curriculum. When a prompt has no description, the description token is masked out of attention. The prediction is read at the query token. The labels and the description are divided by $\sqrt{a_0}$ before they enter the network. Gradients are clipped to norm 1, and training uses bfloat16 autocast. The number of demonstrations in a training prompt is uniform on 0 to 20 (0 to 15 for the $r = 0.05$ and $r = 0.5$ networks described below).

Four further networks were trained once each, at $r = 0.5$ ($p = 1$ and $p = 0.9$) and at $r = 0.05$ ($p = 1$ and $p = 0.9$). Each reaches an ESS within 0.03 of the Bayes-optimal ESS. The four values are 2.64 compared with 2.66, 2.16 compared with 2.15, 7.32 compared with 7.35, and 4.38 compared with 4.37.

Table 6.1: **The meta-trained networks against the Bayes-optimal predictors.** Here $d = 5$ and $a_0 = 10$, and each network is evaluated on 20,000 prompts. Rows are grouped by training reliability, and "Seeds" gives the number of networks, trained with different random seeds, behind each row. "Network" is the mean ESS over these networks, ± the standard deviation across them; for a single network, it is ± the standard error over prompts. "Trained" and "calibrated" are the Bayes-optimal predictors that use the training and the test reliability, respectively, and they are shown once where the two coincide. Calibrated values, and trained values where the two reliabilities coincide, use 200,000 prompts (standard errors at most 0.1). Other trained values use the network's evaluation prompts. Prompt noise alone can move these values by up to 0.6, so a trained value can exceed the calibrated one. "Trust" is the network's implied trust with no demonstrations, averaged over the networks in the row.

				ESS			
r	p_{train}	p_{test}	Seeds	Network	Trained	Calibrated	Trust
0.1	1	1	2	5.85 ± 0.03	5.89	—	0.964
0.1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
0.1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
0.01	1	1	3	14.26 ± 0.48	15.91	—	1.008
0.01	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
0.01	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
0.1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
0.1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
0.1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
0.01	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
0.01	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
0.01	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
0.1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
0.1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
0.1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
0.01	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
0.01	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
0.01	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

Chapter 7

Pretrained Language Models

The networks of Chapter 6 learn a description's specificity and reliability from their training data. A pretrained language model can learn these properties only from the prompt. In this chapter, we therefore state both properties in the prompt, together with the noise level. We then compare the description's ESS for the model with its ESS for a predictor that is Bayes-optimal given what the prompt states. The design follows Gupta et al. (2025) and Akata et al. (2026). A stated prior is a description whose Bayes-optimal ESS is known exactly, so the model is measured against a known optimum.

7.1 Setting

We use the simplest case of the setting of Chapter 3, with one weight ($d = 1$) and a fixed input $x = 1$. A task is then a hidden mean $w \sim \mathcal{N}(500, 100^2)$, and a demonstration is a number $y \sim \mathcal{N}(w, \sigma^2)$, rounded to an integer. A description states a value m and a standard deviation τ, and tasks are drawn so that $w \mid m \sim \mathcal{N}(m, \tau^2)$. This is the description of Section 3.2 with $r = \tau^2/100^2$. We set the noise to $\sigma = 60$ and τ to 60, 30, or 10, and we show $n = 0$, 1, 2, 4, 8, 16, 32, or 64 demonstrations. A second noise level, $\sigma = 30$ with $\tau = 30$, 15, or 5, serves as a check.

An *irrelevant* description states the value that a relevant description of an independently drawn task would state. It is therefore unrelated to the task, though not necessarily far from it. A *stated reliability* adds a sentence that gives a 10% or 50% chance that the description is wrong. This is the mixture of Section 3.2 with $p = 0.9$

or 0.5, expressed in words. The stated value and the demonstrations stay the same as without the sentence.

In this design, each τ comes with its own stated value, drawn so that it is accurate to τ. A more specific description therefore also states a value closer to the truth. A *specificity control* separates the two. It states the value drawn for $\tau = 10$, or the value drawn for $\tau = 60$, together with each of $\tau = 60$, 30, and 10, on the same tasks and demonstrations. Only the stated specificity then changes (Section 7.6.1).

Prompt. The prompt is the user turn of the model's chat template. The description is a single sentence of it, as in this example:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.

548
431

Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is replaced by "Its mean is unknown." With a stated reliability, a third sentence follows: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." With no demonstrations, the question is "Predict the first number." The stated value of a relevant description is drawn given the task, so that $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. In the rare case (under 1%) that a number or a stated value falls outside the range 100 to 999, the task is redrawn.

7.2 The Bayes-Optimal Predictor

The Bayes-optimal predictor in this chapter knows only what the prompt says: the noise level σ and the description. Beyond that, the mean is unknown, and we model it as uniform over the three-digit numbers, 100 to 999. This predictor is optimal for what the prompt states rather than for the way the tasks are actually drawn, and we call it Bayes-optimal in that sense. Without a description, its prediction is essentially the mean of the demonstrations shown, since the range barely truncates it. Given a relevant description and n demonstrations with mean $\bar{y}$, it predicts $(c\,m + n\,\bar{y})/(c + n)$ with $c = \sigma^2/\tau^2$, so it puts a weight of $c/(c + n)$ on the stated value. This is the conjugate normal model of Section 2.5 with $n_0 = c$. The description adds a precision

of $1/\tau^2 = c/\sigma^2$, equal to that of c demonstrations. The squared error is therefore $\sigma^2/(c+n)$ with the description and σ^2/n without it, and the description's Bayes-optimal ESS is c at every n, up to the truncation of the uniform prior. Unless told otherwise, the predictor takes every description as relevant. An irrelevant description therefore has a negative Bayes-optimal ESS here, and unlike in Chapters 4 to 6, a model can do better than this predictor. Our values of τ give $c = 1$, 4, and 36 at both noise levels, and we refer to the descriptions by their value of c.

Under the prior from which the tasks are actually drawn, $\mathcal{N}(500, 100^2)$, a predictor without the description would hold a prior precision worth $\sigma^2/100^2 = 0.36$ demonstrations at $\sigma = 60$ (0.09 at $\sigma = 30$). A relevant description's exact ESS would then be $c - 0.36$, that is, 0.64, 3.64, and 35.64. This is the $(1 - r)\,c$ of Chapter 4. We nevertheless keep the predictor that knows only what the prompt states, since the models are told nothing else.

7.3 Models

We use six open chat models. Three of them, gpt-oss-20b (OpenAI, 2025), Phi-4 (14B; Abdin et al., 2024), and OLMo 2 32B Instruct (Team OLMo et al., 2025), encode every three-digit integer as a single token. For these models, the distribution of the reply's first token, restricted to the 900 number tokens and renormalized, is the model's distribution over the next number, and we take its mean as the model's prediction. The other three, Qwen3.5 9B (Qwen Team, 2026), Gemma 4 12B, and Gemma 4 31B (Gemma Team, 2026), split numbers into digits. For these models, the prediction is the number they write under greedy decoding. On gpt-oss-20b, the two readouts give weights within 0.21 of each other (Section 7.6.3). Thinking is disabled where the chat template allows it, and we start gpt-oss-20b's reply in its final channel. The models run in Hugging Face Transformers at their native precision (`openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`), and the tasks come from a seeded generator.

We also tested three other models but do not report them, because they begin too few of their replies with a number. MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct begin only 62%, 20%, and 10% of their replies with a number. They often explain instead ("To predict…", "Given…"), despite the

instruction to reply with the number only. For example, OLMo 2 7B's ESS for the $c = 36$ description reads 21, but this value comes from a readout that rests on a minority of the model's probability. We report a model only if it begins at least three quarters of its replies with a number overall, and at least half of its no-description replies at $n = 1$, 2, and 4, the curve from which its ESS is read.

Two further sets of runs use models that reason before answering. Section 7.6.6 runs gpt-oss-20b with reasoning turned on, and Claude Haiku 5.5. Section 7.6.7 runs a frontier model, GPT-5.6-Luna, inside an agent harness.

7.4 Measurement

This section defines how we read a model's ESS and its weight on the stated value. A model's error is the root-mean-square distance between its prediction and the hidden mean over 2,000 tasks. The prompt asks for a number rather than a distribution, so we apply Definition 3.1 with squared error. We read the ESS from the model's own no-description error curve. The no-description mean squared error is known at $n = 1, 2, 4, \ldots, 64$. Between these points, we interpolate it linearly in $\log n$, on the squared error rather than its root. The ESS ESS_n is the first t at which this curve falls to the description's mean squared error at n, minus n. This first crossing is the n^* of Definition 3.1, the smallest t at which the two errors are equal. The curve starts at one demonstration. When one demonstration alone beats the description, we therefore report the bound $\mathrm{ESS}_n \leq 1 - n$. When the description's error is below the no-description curve at every t measured, up to 64, we report the bound $\mathrm{ESS}_n > 64 - n$. Figure 7.1 shows the error curves from which the ESS is read.

We summarize the ESS by its median over $n = 1$, 2, and 4, with 95% bootstrap intervals over tasks (Table 7.1). Appendix A gives every n. The summary leaves out $n = 0$, where the models rarely answer with a number when no description is given. It also leaves out large n, where the ESS mixes the description's value with two other effects. The first is the models' slow learning from demonstrations. After 64 demonstrations at $\sigma = 60$, their error is 9 to 23, compared with 7.5 for the Bayes-optimal predictor. The second is their over-reliance on loose descriptions. A model that learns slowly from demonstrations gives a description a higher ESS, because each of its own demonstrations is worth less. The weights defined below do not depend on this. Each bootstrap replicate resamples tasks, using the same tasks for both curves,

and repeats the reading. The intervals therefore reflect sampling over tasks under this procedure, but not the interpolation bias that we quantify next.

Computed in the same way, the Bayes-optimal ESS is 1.3, 4.4, and 36 rather than 1, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Two effects cause these differences. The $\log n$ interpolation biases the ESS upward, by about 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36 when computed with exact errors. The finite sample of tasks shifts it further, in either direction. For $c = 36$, it moves the value down to 36 at $\sigma = 60$ and up to 43 at $\sigma = 30$. The tables report the value computed in this way for the Bayes-optimal predictor and the models alike. Section 7.6.2 compares this reading of the curves with three others.

A model's own no-description curve can rise with n. This happens with some models that reason, which make occasional large errors when they average many numbers by hand. The first crossing of such a curve can lie beyond the grid, and the ESS read from it then says little. Where a model's own no-description curve rises, we therefore also read the *reference ESS*. It is read in the same way, but off the Bayes-optimal predictor's no-description curve instead of the model's own. The reference ESS is the number of demonstrations that a Bayes-optimal learner without the description would need to match the model's error with the description.

We also measure a model's *weight* on the stated value. This is the coefficient on m in a regression, across tasks, of the model's prediction on m and on the demonstrations' mean $\bar{y}$, with a constant. For the Bayes-optimal predictor, the weight is $c/(c+n)$. A reply without a number is left out of both the error curves and the weight regression. For a model read from its distribution, the share of replies that begin with a number is the probability that its first token puts on the number tokens. For a model read from its written answer, it is the share of replies that contain a number. Where this share is below three quarters, Tables 7.5 and A.3 set the cell in italics.

7.5 Results

A specific description is worth a small fraction of its Bayes-optimal ESS. The $c = 36$ description puts the standard deviation of the mean at 10. Across the six models, it is worth a median of 1.4 to 7.6 demonstrations, compared with a Bayes-optimal ESS of 36, and at most 15 at any of $n = 1$, 2, and 4 (Figure 7.2a; Table 7.1). The $c = 1$ and $c = 4$ descriptions are worth 0.0 to 1.9 and 0.2 to 4.1

Table 7.1: **The ESS and the weight on the stated value for each model.** Left: the ESS of a description, as the median over $n = 1$, 2, and 4 demonstrations shown. Columns give a relevant description with $c = 1$, 4, or 36, and the $c = 36$ description when irrelevant or with a stated 10% or 50% chance of being wrong. Brackets give 95% bootstrap intervals, and $\leq$ marks an upper bound, used where one demonstration alone beats the description. Here $\sigma = 60$ and there are 2,000 tasks; "written" marks a model read from the number it writes. Right: the weight on the stated value with one demonstration shown. The Bayes-optimal weight is $c/(c + n)$. The Bayes-optimal predictor takes a description as relevant unless a chance of being wrong is stated, in which case it hedges with that chance.

	ESS, median over $n = 1$, 2, 4						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	stated 10%	stated 50%	$c = 1$	$c = 4$	$c = 36$
Bayes-optimal	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

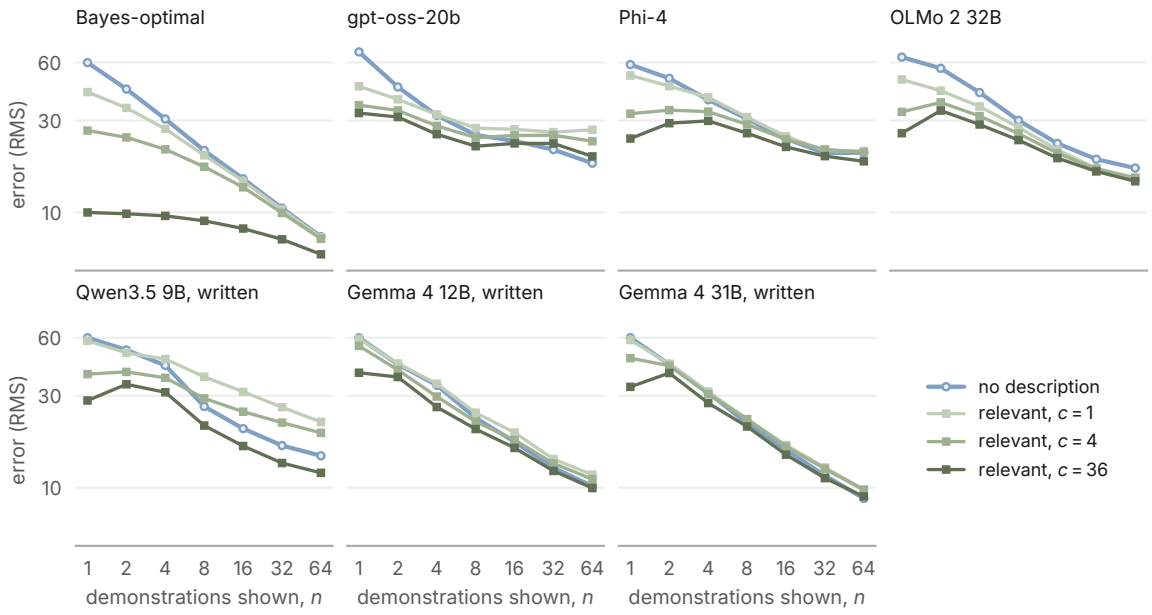


Figure 7.1: **The error curves from which the language-model ESS is read.** Each panel shows the root-mean-square error against the number of demonstrations shown, without a description (blue) and with a relevant description (green, darker for larger c). The panels show the Bayes-optimal predictor and the six reported models. Here $\sigma = 60$ and there are 2,000 tasks, and replies without a number are left out. The ESS of a description at n is the horizontal distance from its point at n to the blue curve.

demonstrations, compared with 1.3 and 4.4. The ESS rises with c mainly because a more specific description states a value closer to the truth. The specificity control shows this directly. With the stated value accurate to $\tau = 10$, gpt-oss-20b's ESS is 2.7, 2.6, and 2.8 whether the prompt states $\tau = 60$, 30, or 10. The Bayes-optimal ESS for these three prompts is 2.5, 13, and 36 (Table 7.2). The conditions with noise $\sigma = 30$, which realize the same three values of c with both standard deviations in the prompt halved, show the same pattern (Table A.1).

The weight follows the stated specificity weakly or not at all. The specificity control measures this directly. Changing only the stated standard deviation from 60 to 10 moves the Bayes-optimal weight by 0.47 at one demonstration. Across $n = 1$, 2, and 4, gpt-oss-20b and OLMo 2 32B move by at most 0.02 of the Bayes-optimal

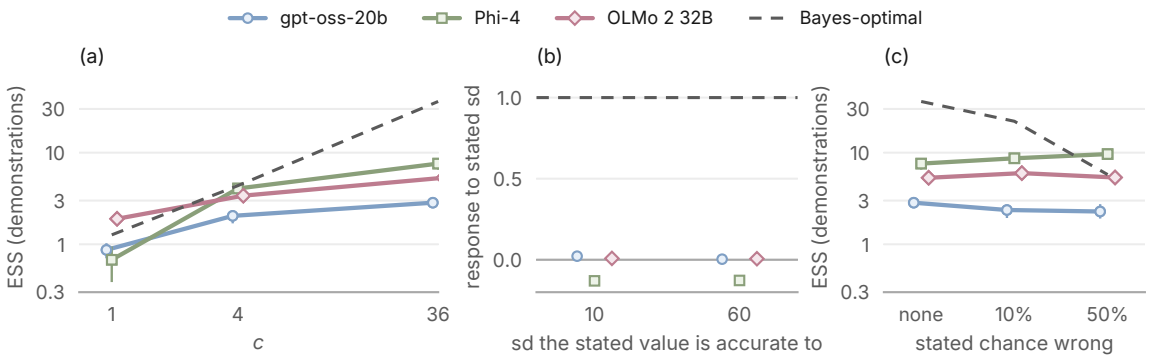


Figure 7.2: **Language models barely adjust to a description's stated specificity or reliability.** The figure shows gpt-oss-20b, Phi-4, and OLMo 2 32B, read from their first-token distribution and answering without reasoning, against the Bayes-optimal predictor (dashed), for a relevant description; Table 7.1 gives the other three models. Here $\sigma = 60$ and there are 2,000 tasks. Error bars show 95% bootstrap intervals over tasks. In (a) and (c), the ESS is the median over $n = 1$, 2, and 4 demonstrations shown. (a) The ESS of the descriptions whose Bayes-optimal ESS is $c = 1$, 4, and 36. (b) The specificity control (Table 7.2). The panel shows the change in prediction when only the stated standard deviation goes from 60 to 10, as a share of the Bayes-optimal change, pooled over $n = 1$, 2, and 4. The stated value is accurate to a standard deviation of 10 or of 60. (c) The ESS of the $c = 36$ description with no stated chance of being wrong, or with a stated 10% or 50% chance. The description is in fact relevant, and the Bayes-optimal predictor takes the stated chance at face value.

change. Phi-4 moves by -0.13 of it, so it relies slightly less on a value stated as more precise. The three models read only from their written answer move more, but still far less than the Bayes-optimal predictor. Qwen3.5 9B moves by 0.12 to 0.16 of the Bayes-optimal change, Gemma 4 12B by 0.12 to 0.19, and Gemma 4 31B by 0.28 to 0.38, mostly at one demonstration (Figure 7.2b; Table 7.2).

The main design shows the same. At $\sigma = 60$, with no demonstrations shown and no stated reliability, the models answer with the stated value (weight 0.92 to 1.10), although Phi-4 and OLMo 2 32B then begin only 20% to 23% and 50% to 59% of their replies with a number. With one demonstration shown, the weight of gpt-oss-20b, Phi-4, and OLMo 2 32B changes by at most 0.08 from $c = 1$ to $c = 36$, and if anything it falls. The Bayes-optimal weight, by contrast, rises from 0.50 to 0.80

to 0.97 (Table 7.1). At every n from 1 to 64 and at both noise levels, the spread of these three models' weights across c is at most 0.16, compared with a Bayes-optimal spread of 0.34 to 0.71 (Table A.3). The weights of the two Gemma models do rise with c, by about 0.25 between $c = 1$ and $c = 36$ at one demonstration (Table 7.1), which is half the Bayes-optimal rise. The weight of Qwen3.5 does not. The models therefore under-use a specific description and, with many demonstrations, over-use the least specific one. At 64 demonstrations, gpt-oss-20b still puts a weight of 0.23 on the $c = 1$ value, compared with the Bayes-optimal 0.02, and that description raises its error (ESS −57).

Models discount a contradicted description. An irrelevant $c = 36$ description has an ESS of at most −0.4, −1.0, and −0.6 for gpt-oss-20b, Phi-4, and OLMo 2 32B, and at most −0.7 to −1.0 for the other three models. These values are bounds, because one demonstration alone beats the description. With more demonstrations, the models discount a stated value that the demonstrations contradict. At $n = 4$, their weight on an irrelevant $c = 36$ value is 0.01 to 0.33, compared with 0.18 to 0.44 on a relevant one. Their errors (35 to 64) therefore stay far below the Bayes-optimal predictor's error of 124, because that predictor takes the description as relevant.

A stated chance of being wrong does not move the ESS in proportion. Stating a 10% or 50% chance that the description is wrong leaves the ESS of gpt-oss-20b and OLMo 2 32B about where it was, and Phi-4's ESS even rises, from 7.6 to 8.7 and 9.7 (Figure 7.2c). Qwen3.5 9B and Gemma 4 31B lower their ESS, but by about as much for a 50% chance as for a 10% chance. Gemma 4 12B's ESS falls for a 10% chance and rises for a 50% chance (Table 7.1). No model hedges in proportion to the stated chance, whereas the Bayes-optimal predictor's ESS falls to 22 and 5.8. We also test actual unreliability, in which one description in ten is irrelevant. The $c = 36$ description is then worth 5.5 demonstrations to the calibrated Bayes-optimal predictor, 0.2 to the fully trusting one, and 0.4 to 3.2 to the models, or −0.1 to 4.0 when the models are told the chance (Section 7.6.4).

With reasoning, the weight follows the stated specificity. We reran these conditions on 200 tasks with two models that reason before answering (Section 7.6.6). The first is gpt-oss-20b at low reasoning effort with up to 4,096 new tokens, so that

99.9% of its replies finish. The second is Claude Haiku 5.5, called through the Claude Code command line. Both models now move with the stated standard deviation as the Bayes-optimal predictor does. In the specificity control, gpt-oss-20b moves by 1.0 of the Bayes-optimal change, and Haiku 5.5 at its default effort moves by 0.92 of it (Table 7.2). Haiku 5.5 also comes close to the Bayes-optimal ESS. Its ESS for the $c = 1$, 4, and 36 descriptions is 1.1, 3.3, and 28, compared with 1.0, 3.5, and 35 for the Bayes-optimal predictor on the same tasks (Table 7.6). It hedges when told of a 50% chance of being wrong, with a weight of 0.55 against the Bayes-optimal 0.58, but not when told of a 10% chance. gpt-oss-20b barely hedges at all. A frontier model, GPT-5.6-Luna, reasoning inside the Codex agent, matches the Bayes-optimal weight to within 0.03. Its ESS for the $c = 36$ description is 32, against a Bayes-optimal 33 on the same 100 tasks (Section 7.6.7). With its reasoning turned off, it shows the same flat weight as the open models.

gpt-oss-20b gets the weight right, but about 2% of its answers are more than 40 from the Bayes-optimal prediction. These large errors come from slips such as swapping the two variances. Its $c = 36$ description therefore has a reference ESS of 24 $[12, 35]$, well below the Bayes-optimal 35. Its own no-description error curve rises after 16 demonstrations, so the ESS read from that curve lies beyond the grid and says nothing. A fitted weight hides such errors, whereas the reference ESS does not.

7.6 Further Analyses

This section gives the analyses behind the results of Section 7.5. It describes the specificity control, other readings of the error curves, the written readout, actual unreliability, the wording of the description, the reasoning runs, and a frontier model inside an agent harness.

7.6.1 Specificity Control

For each task, the control states the value drawn for $\tau = 10$, or the value drawn for $\tau = 60$, together with each of $\tau = 60$, 30, and 10. It uses the same 2,000 tasks and demonstrations as the main run, together with its own no-description condition, at every n (Table 7.2). Some of its cells repeat cells of the main run, because they state a value with the standard deviation for which it was drawn. These repeated cells

reproduce the main run exactly for every model except gpt-oss-20b. For gpt-oss-20b, the distribution readout differs by a median of 0.8 to 1.2 (at most 39), and the written answers differ in 12% of replies (median difference 0). The likely cause is a newer library version.

The primary contrast is the paired change in each task's prediction when the stated standard deviation goes from 60 to 10. We regress this change, across tasks and $n = 1$, 2, and 4, on the change in the Bayes-optimal prediction, using the tasks answered in both cells. The coefficient is 1 for a learner that follows the stated specificity as the Bayes-optimal predictor does, and 0 for one that ignores it. The Bayes-optimal reference takes the statement at face value. With the value accurate to $\tau = 60$ but stated as $\tau = 10$, its ESS is therefore about -0.8, and a model that ignores the stated specificity can beat it there.

With the stated value fixed, the accuracy of that value, not the stated specificity, mainly sets the ESS of the models answering without reasoning. With the value accurate to $\tau = 60$, each one's ESS is between -0.1 and 2.0 under every stated τ. With the value accurate to $\tau = 10$, each one's ESS is higher. From a stated $\tau = 60$ to a stated $\tau = 10$, however, it rises by at most 1.2, whereas the Bayes-optimal ESS rises from 2.5 to 36. Phi-4's ESS even falls, from 12 under a stated $\tau = 60$ to 7.6 under a stated $\tau = 10$, the reverse of the Bayes-optimal order. The models that reason respond to the stated specificity much as the Bayes-optimal predictor does. gpt-oss-20b at low effort responds by 1.03 and 1.00 of the Bayes-optimal change, and Haiku 5.5 by 0.92 at medium effort and by 0.70 and 0.79 at low effort.

7.6.2 Readings of the Error Curves

Definition 3.1 interpolates between the numbers of demonstrations at which the error is known. The grid here doubles n, so how the curve is read between grid points matters. On the Bayes-optimal predictor's exact errors, σ^2/n without the description and $\sigma^2/(c+n)$ with it, interpolation in $\log n$ overstates the ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Interpolation in n overstates it by 0.3, 0.6, and 4.1. For models answering without reasoning, the first crossing is also the only one. Of their twelve no-description curves at the two noise levels, only Phi-4's at $\sigma = 60$ rises, from 20.3 at 32 demonstrations to 20.4 at 64.

Table 7.3 gives the headline ESS under four readings of the no-description curve.

Table 7.2: **With the stated value and the demonstrations fixed, models answering without reasoning respond to the stated specificity far less than the Bayes-optimal predictor, and models that reason follow it.** Specificity control at $\sigma = 60$. The two left columns give the response to the stated standard deviation. This is the paired change in prediction from a stated standard deviation of 60 to one of 10, as a share of the Bayes-optimal change, over $n = 1$, 2, and 4, with 95% bootstrap intervals, for a value accurate to $\tau = 10$ or to $\tau = 60$. The right columns are for the value accurate to $\tau = 10$. They give the weight on that value at $n = 1$ and the ESS (median over $n = 1$, 2, and 4) under each stated τ. The Bayes-optimal predictor takes the stated τ at face value. The upper rows use 2,000 tasks and are read as in Table 7.1. The lower rows are the reasoning runs on the first 200 tasks, on which the Bayes-optimal ESS is 2.2, 14, and 35.

	response to the stated sd		value accurate to sd 10, stated sd 60 / 30 / 10	
	value sd 10	value sd 60	weight, $n = 1$	ESS
Bayes-optimal	1	1	0.50 0.80 0.97 2.5 13	36
gpt-oss-20b	0.02 [0.02, 0.03]	0.00 [0.00, 0.01]	0.48 0.47 0.49 2.7 2.6	2.8
Phi-4	−0.13 [−0.14, −0.12]	−0.13 [−0.13, −0.12]	0.81 0.75 0.74 12 7.2	7.6
OLMo 2 32B	0.01 [0.01, 0.01]	0.01 [0.00, 0.01]	0.70 0.70 0.70 4.8 4.6	5.3
gpt-oss-20b (written)	0.01 [−0.01, 0.04]	−0.01 [−0.03, 0.01]	0.52 0.46 0.50 2.9 3.5	3.8
Qwen3.5 9B (written)	0.12 [0.09, 0.14]	0.16 [0.13, 0.18]	0.76 0.75 0.80 3.3 3.5	4.0
Gemma 4 12B (written)	0.19 [0.16, 0.23]	0.12 [0.09, 0.14]	0.38 0.29 0.59 1.0 1.0	1.8
Gemma 4 31B (written)	0.38 [0.34, 0.41]	0.28 [0.25, 0.31]	0.43 0.49 0.72 0.2 0.3	1.4
gpt-oss-20b, low	1.03 [0.93, 1.14]	1.00 [0.96, 1.04]	0.50 0.79 0.95 0.5 8.6	> 60
Haiku 5.5, medium	0.92 [0.84, 0.99]	0.92 [0.86, 0.97]	0.36 0.69 0.74 2.1 11	28
Haiku 5.5, low	0.70 [0.59, 0.79]	0.79 [0.70, 0.86]	0.30 0.34 0.41 2.4 16	26

The first is ours. The second interpolates in n instead of $\log n$. The third first makes the curve non-increasing by a least-squares monotone fit (pool adjacent violators). The fourth fits the curve $A/(n+\kappa)+B$ to the no-description errors by relative least squares. This is the form of a Bayesian learner's error plus a floor, and it recovers the exact ESS from the Bayes-optimal predictor's exact errors. No model's headline ESS moves by more than 1.4 demonstrations across the four readings. None exceeds 8.1 for the $c = 36$ description, against 35 to 38 for the Bayes-optimal predictor.

The readings differ more at larger n, where a model's no-description curve flattens and a small difference in error spans many demonstrations. For example, gpt-oss-20b's $c = 36$ ESS at $n = 8$ is 17 in Table A.1 and 9.2 from the fitted curve. In Figure 7.1, the error of four models with the $c = 36$ description rises from one demonstration shown to two. Each demonstration pulls their prediction off the stated value much more than it pulls the Bayes-optimal prediction (Table A.3).

Table 7.3: **The headline ESS under four readings of the no-description curve.** Each entry is the median over $n = 1$, 2, and 4 demonstrations shown for one of the three relevant descriptions; $\sigma = 60$ and there are 2,000 tasks. "Log n" is the reading of Table 7.1, the first crossing with interpolation in $\log n$. "Linear" interpolates in n, "monotone" reads the curve after a monotone fit, and "fit" reads the fitted curve $A/(n+\kappa)+B$. The exact Bayes-optimal ESS is 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Bayes-optimal	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

7.6.3 Written Readout

For Qwen3.5 9B, Gemma 4 12B, and Gemma 4 31B, and as a check on the other models, the prediction is read from the reply under greedy decoding with eight new tokens. It is the first number in the reply with three digits before any decimal part, with the decimals kept. This is the rule for every text reply in this chapter. The written runs of the open models were scored before the rule kept decimals, so they

store only the integer part of each answer. This matters little, because a model that answers directly rarely writes a decimal part: only 3 of GPT-5.6-Luna's 1,300 replies with reasoning off have one. A reply without such a number is left out of both the error curves and the weight regression. The share of replies that contain one is reported as "number" in the tables.

On gpt-oss-20b, the written readout gives weights within 0.21 of the distribution readout's in every condition and at every n. Its headline ESS medians are 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution. The written readout is the noisier of the two, since a single greedy answer stands in for the mean of the distribution. Table A.2 gives the written readout's ESS at every n for the three models read only from their written answer.

Phi-4 begins only 63% of its written replies with a number over all conditions, so Table 7.1 reads it from its distribution only. In the wording ablation of Section 7.6.5, however, its share is above three quarters for the wording of Section 7.1. OLMo 2 32B writes a number in only 5%, 36%, and 48% of its no-description replies at $n = 1$, 2, and 4 ($\sigma = 60$), so we do not report its written ESS. Its written weights on the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with 0.72, 0.70, and 0.70 from its distribution. Its first token puts 41% of its probability on numbers in the no-description prompts with one demonstration shown, and 75% with two. Its no-description prediction at $n = 1$ therefore rests on a minority of that probability.

7.6.4 Actual Unreliability

The stated-reliability conditions keep the description relevant and test whether a model hedges as it is told to. Table 7.4 instead realizes the mixture of Section 3.2. On the same tasks, the squared errors of the relevant and the irrelevant conditions are combined with weights p and $1 - p$ before the ESS is read. We do this both for prompts that do not state the chance and for prompts that do. We compare the models with two Bayes-optimal predictors: the fully trusting one and the calibrated one, which hedges at p. With one description in ten irrelevant, the $c = 36$ description is worth 5.5 demonstrations to the calibrated predictor and 0.2 to the fully trusting one. To the models, it is worth 0.4 to 3.2 demonstrations when they are not told the chance and -0.1 to 4.0 when they are. Their flat weight on a description thus acts as a hedge between full trust and calibration, and being told the chance moves their

ESS by at most two demonstrations, in either direction. With half of the descriptions irrelevant, the calibrated predictor keeps less than one demonstration, and no model keeps more than one.

Table 7.4: **The ESS under actual unreliability.** The relevant and irrelevant conditions are combined with weights p and $1 - p$. Each entry is the median over $n = 1$, 2, and 4 demonstrations shown, with 95% bootstrap intervals over tasks; $\sigma = 60$ and there are 2,000 tasks. An entry is a bound where one demonstration alone beats the description at some n.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Bayes-optimal, takes it as relevant	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Bayes-optimal, hedges at p	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	-0.1 [-0.1, 0.0]	-0.1 [-0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	-1.0 [-1.0, -0.7]	-0.1 [-0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

7.6.5 Wording of the Description

We reran the three relevant descriptions and the no-description prompt at $n = 1$, 2, and 4 with three other wordings of the description sentence: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." These runs use the written readout and 2,000 tasks (Table 7.5). The wording sets how much a model relies on the description. With one demonstration shown, Gemma 4 12B's weight on the $c = 1$ value is 0.34 under the wording of Section 7.1 and 1.00 under each of the other three. For gpt-oss-20b, the same weight ranges from 0.17 to 0.67. (A weight above 1 means that the prediction moves more than the stated value does.) Under no wording, however, does the weight follow the stated specificity as the Bayes-optimal weight does. Over all models and wordings, the gap between the weights for $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$

and 0.44 at $n = 4$, compared with 0.47 and 0.70 for the Bayes-optimal predictor.

Table 7.5: **The wording of the description sets how much a model relies on it, but under no wording does the weight follow the stated specificity.** The weight on the stated value for the three relevant descriptions at $n = 1$ and 4 under four wordings (written readout, 2,000 tasks). The wording marked "the paper's" is the wording of Section 7.1. "Number" is the share of replies that contain a number, and italics mark cells where this share is below three quarters. The Bayes-optimal weight is 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (the paper's)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (the paper's)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (the paper's)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (the paper's)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (the paper's)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (the paper's)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

7.6.6 Reasoning

Both reasoning runs use the first 200 tasks of the main run. They measure the no-description curve at $n = 1$ to 64, and each description and control condition at $n = 1$, 2, and 4 (Tables 7.2 and 7.6). For Haiku 5.5, the no-description curve also includes $n = 40$ and 48, near where its $c = 36$ ESS is read.

gpt-oss-20b runs in vLLM with greedy decoding at low reasoning effort and up to 4,096 new tokens. Of its 7,400 replies, 99.9% reach the final channel, and at least 97% do in every cell. The replies use 322 tokens on average. The number in the final channel, read by the rule of the written readout, is the model's prediction. An earlier run with 384 tokens finished only 28% to 54% of the replies where a reliability is stated. The hedge between a 10% and a 50% chance that this earlier run seemed to show came from which replies finished.

Claude Haiku 5.5 (`claude-haiku-5-5`) is called through the Claude Code command line on a subscription, with one prompt per call. Each call has no tools, no project or user instructions, and a one-line system prompt. We ran it at its default (medium) reasoning effort and at low effort, and its replies are sampled at the command line's default temperature. At medium effort, it thinks before most answers, with 270 thinking tokens on average. It thinks in 37% to 88% of replies at $n = 1$ over all conditions, and in nearly all replies from $n = 2$. All but one of its 7,800 replies at medium effort contain a number, and all but three at low effort. Its no-description error is close to the Bayes-optimal predictor's through $n = 48$. At $n = 16$, for example, it is 15.8 against 14.8, and two replies there are far off. Its error rises at $n = 64$, where in 9 of 200 replies it divides the sum by 60 instead of 64.

Within each cell, Haiku 5.5's replies split by whether it thinks. At one demonstration, the replies that think put weights of 0.50, 0.80, and 0.97 on the relevant $c = 1$, 4, and 36 values. These are the Bayes-optimal weights to two decimals, at either effort. The replies that answer directly put a weight of 0.00 to 0.05 on the stated value and repeat the number shown. Its overall weight at one demonstration is a mixture of the two kinds of reply, 0.44, 0.66, and 0.74 at medium effort (Table 7.6). On these three descriptions, it thinks in 34% to 53% of replies at low effort, against 72% to 82% at medium effort. Its overall weight at one demonstration is therefore lower at low effort and nearly flat, at 0.36, 0.34, and 0.41. With more demonstrations, it thinks more often, and its weight is close to the Bayes-optimal weight again. At four demonstrations, it is 0.17, 0.48, and 0.86, against 0.20, 0.50, and 0.90. The choice to think is the model's own, so this split describes its replies rather than the result of an experiment.

The columns "ref." of Table 7.6 give the reference ESS of Section 7.4. They read the model's error with the description off the Bayes-optimal no-description curve instead of the model's own. For gpt-oss-20b, whose own curve rises after 16 demonstrations,

the ESS of the $c = 36$ description read from its own curve lies beyond the grid, while its reference ESS is 24.

Table 7.6: **With reasoning, Haiku 5.5 comes close to the Bayes-optimal ESS, while gpt-oss-20b gets the weight right but loses ESS to a few large errors.** The runs use the first 200 tasks at $\sigma = 60$. The table gives the weight on the stated value at $n = 1$ and the ESS as the median over $n = 1$, 2, and 4, with 95% bootstrap intervals. The ESS is read from the model's own no-description curve, and the reference ESS ("ref.") from the Bayes-optimal predictor's. The symbol $\leq$ marks an upper bound, and > 60 marks an ESS beyond the grid. The Bayes-optimal values are read in the same way on the same tasks.

	Bayes-optimal		gpt-oss-20b, low			Haiku 5.5, medium			Haiku 5.5, low		
Description	weight	ESS	weight	ESS	ref.	weight	ESS	ref.	weight	ESS	ref.
$c = 1$	0.50	1.0	0.50	0.3 [−0.2, 1.0]	0.3 [−0.2, 1.0]	0.44	1.1 [0.6, 1.6]	1.1 [0.6, 1.6]	0.36	0.8 [0.5, 1.7]	0.7 [0.4, 1.2]
$c = 4$	0.80	3.5	0.78	1.0 [−0.1, 3.1]	1.0 [−0.1, 3.1]	0.66	3.3 [2.2, 4.7]	3.5 [2.2, 4.8]	0.34	3.4 [2.5, 5.0]	3.3 [2.1, 4.5]
$c = 36$	0.97	35	0.95	> 60 [13, >]	24 [12, 35]	0.74	28 [21, >]	28 [21, 39]	0.41	26 [22, 32]	24 [17, 31]
irrelevant	0.97	≤ -1.0	0.97	≤ -1.0	≤ -1.0	0.87	≤ -1.0	≤ -1.0	0.60	≤ -1.0	≤ -1.0
stated 10%	0.88	27	0.95	1.3 [−0.2, >]	1.2 [−0.3, 15]	0.79	33 [26, >]	33 [26, 40]	0.42	27 [22, 38]	26 [20, 37]
stated 50%	0.58	5.7	0.89	8.8 [4.2, >]	8.8 [4.2, 17]	0.55	3.8 [2.3, 6.4]	3.9 [2.4, 6.5]	0.49	5.7 [4.3, 21]	5.6 [4.2, 9.2]

7.6.7 A Frontier Model Inside an Agent Harness

The models above are open models run locally, so that their first-token distributions can be read. The squared-error ESS, however, needs only the models' answers. We therefore also tested GPT-5.6-Luna (model `gpt-5.6-luna`) through the Codex command-line agent, running in its read-only sandbox on a subscription. We gave it the prompts for the three relevant descriptions ($c = 1$, 4, and 36; $\sigma = 60$) with $n = 1$ and 4 demonstrations, on 100 tasks each. We ran each prompt twice: once with the reasoning effort set to low (about 115 reasoning tokens per answer) and once with reasoning turned off. This measures the model as it is deployed in the agent. The agent's instructions and tool definitions, about 26,000 tokens, surround the prompt, much as a provider's system prompt surrounds a prompt sent through an API. The model used no tool in any of these 1,200 replies, and every reply is a number alone. Ten replies have a trailing period, and 181 have a decimal part, which we keep.

With reasoning, the model's weight on the stated value matches the Bayes-optimal weight $c/(c+n)$ to within 0.03 in all six cells. Let a and b be the weights on the stated value and on the demonstrations' mean. The implied worth of the description, $n\,a/b$,

is then c to within half a demonstration (Table 7.7), and the model's error matches the Bayes-optimal predictor's to within 0.8. The prompt therefore contains what is needed to reach the Bayes-optimal weight, and this model reaches it when it reasons. With reasoning turned off, the same model shows the flat weight of Section 7.5. Its weight is 0.72 to 0.88 at $n = 1$ and 0.40 to 0.47 at $n = 4$, whatever the stated specificity, and the implied worth is 2.6 to 5.8 demonstrations for descriptions worth 1 to 36.

To read the ESS of Definition 3.1 from this model's own no-description curve, we also sent the no-description prompts on the same 100 tasks at $n = 1, 2, 4, \ldots, 64$ in both modes. All 1,400 of these replies are numbers. With reasoning, the model averaged the shown numbers with a shell command in the agent's sandbox in 174 of these 700 prompts (4 at $n = 16$, 74 at $n = 32$, and 96 at $n = 64$). It used Python in 127 of these prompts and Node.js or awk in the rest. Its no-description error then matches the Bayes-optimal predictor's to within 0.1 at every n. For example, it is 9.4 and 7.0 at $n = 32$ and 64, compared with 9.4 and 6.9. With reasoning turned off, it used no tool. A more accurate no-description curve lowers the ESS read from it, so the tool use does not inflate the ESS. With reasoning, the ESS of the three descriptions at $n = 1$ and 4 matches the Bayes-optimal predictor's, read in the same way from its predictions on the same tasks. The two agree to within 0.2 demonstrations for $c = 1$ and 4, and give 32 and 33 for $c = 36$ (Table 7.8). With only 100 tasks, the reading procedure itself moves the Bayes-optimal value of 4 to 2.4 and 2.9. With reasoning turned off, the $c = 36$ description is worth 5.4 demonstrations at $n = 1$ and 1.3 at $n = 4$, which is in the range of the open models of Section 7.5, and the $c = 1$ description hurts at $n = 4$.

Table 7.7: **A frontier model reaches the Bayes-optimal weight when it reasons, and shows the flat weight of the open models when it does not.** GPT-5.6-Luna runs inside the Codex agent harness, with 100 tasks per cell and relevant descriptions at $\sigma = 60$. The table gives the weight on the stated value and the implied worth $n\,a/b$, from the regression of Section 7.4. Bootstrap standard errors of the weights are at most 0.03 with reasoning and 0.07 without.

	weight on the stated value						implied worth, demonstrations					
	$n = 1$			$n = 4$			$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36	1	4	36	1	4	36
Bayes-optimal	0.50	0.80	0.97	0.20	0.50	0.90	1	4	36	1	4	36
GPT-5.6-Luna, low reasoning	0.50	0.77	0.97	0.21	0.50	0.90	1.0	3.6	35.7	1.0	4.0	36.5
GPT-5.6-Luna, reasoning off	0.79	0.72	0.88	0.43	0.40	0.47	3.6	2.6	5.8	3.1	2.7	3.7

Table 7.8: **When it reasons, GPT-5.6-Luna's ESS is consistent with the Bayes-optimal ESS.** The ESS at $n = 1$ and 4 demonstrations shown, read as in Section 7.4 from the model's own no-description curve ($n = 1$ to 64, on the same 100 tasks), with 95% bootstrap intervals over tasks. With reasoning, the model averaged the numbers with a shell command in 174 of the 700 no-description prompts, mostly at $n = 32$ and 64. The Bayes-optimal predictor's predictions on the same tasks are read in the same way.

	$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36
Bayes-optimal, read the same way	1.0 [0.6, 1.8]	2.4 [1.6, 3.9]	33 [25, 49]	1.7 [0.5, 3.6]	2.9 [1.2, 7.7]	33 [23, 50]
GPT-5.6-Luna, low reasoning	1.1 [0.6, 1.8]	2.3 [1.4, 3.8]	32 [25, 48]	1.5 [0.3, 3.4]	3.0 [1.2, 7.7]	32 [23, 49]
GPT-5.6-Luna, reasoning off	0.1 [0.0, 0.5]	1.2 [0.8, 2.1]	5.4 [2.5, 14]	−2.2 [−2.6, −1.4]	0.1 [−0.7, 2.5]	1.3 [−0.2, 3.8]

Chapter 8

Conclusion

This thesis asked how many demonstrations a description is worth, and what properties of the description determine that number. For the Bayes-optimal predictor in in-context linear regression, the ESS of a reliable description grows with the task dimension and with the description's specificity. A description of only part of the task is worth about the number of directions it covers, until demonstrations resolve the rest. A description that may be irrelevant behaves differently. With no demonstrations in hand, its ESS is bounded however specific it is. Once demonstrations are in hand, a learner that trusts it fully can do worse than one that ignores it. Networks trained on the task distribution come close to the Bayes-optimal ESS, and they learn how much to trust a description from their training data. Pretrained language models answering without reasoning respond to a description's stated specificity far less than the Bayes-optimal predictor does, even when nothing else in the prompt changes. Models that reason before answering follow the stated specificity, and two of them come close to the Bayes-optimal ESS. A few large errors can still cost a reasoning model much of its ESS while leaving its fitted weight on the description intact. Because the squared-error ESS needs only a learner's answers, the same question can be put to any model and to richer descriptions.

8.1 Limitations

The analysis assumes a linear-Gaussian task and uses the robust mixture as the process that generates descriptions. Under this model, a relevant description errs only within

its stated spread, and an irrelevant one carries no information. A real description, by contrast, can be partly right or more confident than it is accurate. The theory covers descriptions of part of the task (Section 4.2), but no experiment in this thesis tests them. The networks are trained at $d = 5$, with three training seeds for the specific description and two for the loose one. The language-model test covers only the simplest case of the setting, a single weight ($d = 1$), in which there is no dimension term. Its description is a stated numeric prior rather than an instruction. Other wordings change how much a model relies on the description, but under no wording does the weight follow the stated specificity (Section 7.6.5). Three of the open models are read from a greedy written answer, so comparisons across models mix differences between models with differences between readouts. Finally, the reasoning runs with gpt-oss-20b and Claude Haiku 5.5 use 200 tasks, and those with GPT-5.6-Luna use 100.

8.2 Future Work

We outline three directions for future work. The first two follow from the limitations above, and the third from the networks of Chapter 6. None of them is established in this thesis.

Testing partial descriptions. Proposition 4.2 predicts that a specific description of part of the task gains value as demonstrations accumulate, while a complete description loses value. At $d = 16$, $a_0 = 10$, and $r = 0.001$, a description of half the directions is worth 8.1 demonstrations with none in hand and 36 with 100 in hand, while a complete description falls from 117 to 109. This thesis computes these values for the Bayes-optimal predictor only. Networks could be trained on descriptions of some of the directions, and their ESS could be read at every number of demonstrations, as in Chapter 6. The language-model test would need several unknown quantities, of which the description states only some. Such a test would also show whether a language model's ESS has a dimension term.

Structural descriptions. The descriptions studied here state where the task lies, through a prior mean with a stated spread. Many instructions instead state what kind of task it is, for example that only three features matter, or that the answer is monotone

in an input. Such a description narrows the hypothesis class rather than pointing at one task. One case of this kind is already covered. A description that confines w to a known s-dimensional subspace states the other $d-s$ directions exactly, so it is a partial description of those directions in the limit $r \to 0$. A heuristic calculation suggests that such a description keeps its value as demonstrations accumulate. Under a flat baseline prior, it would lower the Bayes-optimal error from about d/n to about s/n, so its ESS would be about $n\,(d/s - 1)$, which grows with the number of demonstrations. Structures that do not fix any direction, such as a sparse task whose relevant features are unknown, or a monotone one, are not covered. The same ESS, networks, and language-model test apply to them. Moreover, a wrong structural description would leave a fully trusting learner with a bias that no number of demonstrations removes.

Where a network stores its trust. The meta-trained networks learn their trust in a description from training (Section 6.2). A natural next step is to locate the components of a network that hold this trust, and to test by intervention whether changing them changes the network's weight on a description without retraining. Such a study would connect the behavioural measure of this thesis to the mechanisms by which networks learn in context.

Bibliography

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Chatlatanagulchai, W., Thonglek, K., Reid, B., Kashiwa, Y., Leelaprute, P., Rungsawang, A., Manaskasemsak, B., and Iida, H. On the use of agentic coding manifests: an empirical study of Claude Code. In *International Conference on Product-Focused Software Process Improvement (PROFES)*, 2025.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Cover, T. M. and Thomas, J. A. *Elements of Information Theory*. Wiley, second edition, 2006.

Evans, M. and Moshonov, H. Checking for prior-data conflict. *Bayesian Analysis*, 1 (4):893–914, 2006.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. *Bayesian Data Analysis*. CRC Press, third edition, 2013.

Gemma Team. Gemma 4. https://huggingface.co/google/gemma-4-31B-it, 2026. Model card.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help transformers learn linear models in-context. In *ICLR*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z. and Lee, K. Dual operating modes of in-context learning. In *ICML*, 2024.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Liu, T. J. B., Boullé, N., Sarfati, R., and Earls, C. J. Density estimation with LLMs: a geometric investigation of in-context learning trajectories. In *ICLR*, 2025.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Reznik, Y. A. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *NeurIPS*, 2017.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

Appendix A

Language-Model Results at Every Number of Demonstrations

This appendix reports the language-model results of Chapter 7 at every number of demonstrations shown. Table A.1 gives the ESS of every description for the Bayes-optimal predictor of Section 7.2 and the three models read from their first-token distribution (Section 7.3). Table A.2 gives the same for the three models read from the number they write (Section 7.6.3). Table A.3 gives the weight on the stated value for the Bayes-optimal predictor and the three models read from their distribution. The ESS is read as in Section 7.4. Tables A.1 and A.3 each continue onto a second page.

Table A.1: The ESS of a description for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution (2,000 tasks). "Number" is the share of a model's replies that begin with a three-digit number, averaged over n, and "10% wrong" and "50% wrong" mark a stated chance of being wrong. The Bayes-optimal predictor is that of Section 7.2, which is optimal for what the prompt states. It takes an irrelevant description with no stated chance of being wrong as relevant, so its ESS can be negative. The symbol $\leq$ marks an upper bound. An entry $> k$ means that the error with the description is below the no-description error at every n measured, up to 64, so the ESS exceeds $k = 64 - n$.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Bayes-optimal	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42

	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table A.2: The ESS of a description for every condition and number of demonstrations shown, for the three models read from the number they write (2,000 tasks). A reply without a number is left out. The columns and the Bayes-optimal values are as in Table A.1.

Predictor	Description	number $n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma=60$, $c=4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma=60$, $c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma=30$, $c=4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma=30$, $c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma=60$, $c=4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma=60$, $c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma=30$, $c=4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma=30$, $c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma=60$, $c=4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma=60$, $c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma=30$, $c=4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma=30$, $c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table A.3: The weight on the stated value for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution. "Number" is defined as in Table A.1. A cell is set in italics where the model begins fewer than three quarters of its replies with a number at that n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60$, $c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60$, $c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60$, $c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60$, $c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60$, $c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60$, $c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60$, $c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60$, $c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60$, $c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60$, $c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60$, $c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60$, $c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60$, $c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30$, $c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30$, $c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30$, $c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60$, $c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60$, $c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60$, $c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60$, $c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60$, $c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60$, $c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30$, $c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30$, $c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30$, $c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17

	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma = 60$, $c = 1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma = 60$, $c = 4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma = 60$, $c = 4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma = 60$, $c = 36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma = 30$, $c = 1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma = 30$, $c = 4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma = 30$, $c = 36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01